Patterns to Principles

Theory Construction in the Social Sciences

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General Information

Course name	From Patterns to Principles: Theory Construction in the Social Sciences
Instruction language	English
Type of instruction	Interactive lectures and group work
Level	Excellence Program TSB
Course load	6 ECTS
Lectures	Dr. C. J. Van Lissa

This course combines literature from empirical social science, methods and statistics, and philosophy to offer a multidisciplinary perspective on theory in the social sciences. There is no textbook: students will read and present on primary literature and discuss it in class. The course uses challenge based learning and a flipped classroom. While default challenges and corresponding materials are available, students are encouraged to bring their own, thereby connecting this course to their other coursework (or future thesis). The course relies on group work and explicitly allows for specialization (i.e., not all students need to master all course-relevant skills).

Learning objectives

This course introduces students to the foundations and applications of theory in social science research. We will analyze the need for better theories based on the current "theory crisis" in empirical social science. We define the characteristics and affordances of "good theories": what do they look like, and how do we use them in (empirical) research? Students will engage with theory construction methods to formalize existing theories or construct their own, and are introduced to formal modeling and simulation studies as tools for theory development and testing. By the end of the course, students will be able to critically assess existing theories and their usage in empirical articles, construct theory, simulate theory-implied data, use theory in empirical research, and relate critically to the state of theory in the social sciences.

Skills

After taking this course, students are able to:

- Remember relevant constructs from the literature
- Understand theory construction methods and their philosophical pitfalls
- Apply theory in empirical research
- Create new - or formalize existing - theory
- Evaluate the use of theory in empirical research

Study Load

This is the breakdown of the expected study load:

Activity	Duration	Times	Total
Synchronous			
Lectures	2	8	16
Tutorials	2	8	16
Asynchronous			
Reading	8	8	64
Assessment type(s)			
Presentation	8	1	8
Writing assignment	2	1	2
Final Report	62	1	62
Total			168
ECTS			6

Assessment

Your grade for the course is based entirely on your "portfolio", comprised of the following components:

Component	Type	Percentage	Description
Multiple choice exam	individual	30.00%	During 8 lecture sessions, a 4-question multip
Writing assignment	individual	10.00%	Short writing assignment (max 500 words), to
Presentation of theory specification	group	20.00%	Present your group project to class
Final theory report	group	40.00%	Write a report about your final theory, and a

Final Theory Report

Your final report should follow the structure of the Theory Paper Format.

The following sections are graded (word limits in parentheses):

1. Introduction and Problem Definition (500 words)
2. Conceptual Outline (200 words)
3. Theory Specification and documentation thereof (no strict word limit; you'll probably use an augmented DAG, PBTS tables, a diagram, or similar)
4. Evaluation of Theory Implications (500 words)
5. *Skip this: Applied example / Case study*
6. Discussion (500 words)

Each of those section will be graded on a 3-point scale (0: unacceptable, 1: good, 2: excellent).

Additionally, we will grade (on the same scale):

- Writing quality
- Independence/creativity (did you just follow along with the course, or did you do something unique?)

The final grade is just a sum of all these grades, divided by the total possible number of points and rescaled to 0-10 point scale.

Group Work

For this course, we want to stimulate you to engage with the material, with one another, and with the teachers. Therefore, some portfolio assignments are made in groups. There are aspects of learning in groups that can really improve your knowledge, like discussion and peer feedback. Groups comprise 3-4 members and are created during the first tutorial session (Section 1.2.1). We encourage students to reflect on their roles in group work, and to explicitly divide roles before each session or switch roles each week. At minimum, it's useful to always have one Facilitator. If your group tends to be unanimous, appoint a Devil's advocate. If your group tends towards disagreement, appoint an Arbitrator.

Course materials

You do not need a book for this course. This GitBook provides all relevant course materials, along with primary literature linked in the GitBook.

Course schedule

Always check the latest course schedule on TimeEdit. For an overview of the content, see below:

Dates	Activities	Topic
21/09 25/09	HC	Why theory needs an update: Theory Crisis
28/09 04/10	LA, Conference	Why theory needs an update: Theory Crisis
05/10 9/10	HC, LA	Why theory needs an update: What Makes a Good Theory?
19/10 23/10	HC, LA	Why theory needs an update: Strategic Ambiguity
26/10 30/10	HC, LA	How to Construct Theory: Latent Theory in Published Papers
02/11 06/11	HC, LA	How to Construct Theory: Formal Theory Specification
9/11 13/11	HC, LA	What is Next: Machine Learning-Informed Theory Construction
16/11 20/11	HC, LA	What is Next: Causal Discovery
23/11 27/11	HC, LA	What is Next: FAIR Theory
30/11 04/12	2xPresentations	Building Theory: Student Theory Workshops (during lecture and tutoria
07//12 11/12	Tentamens blok 2	

Communication

Communication about the course occurs through Canvas (Login with your student ID and password).

Staff

Coordinator/lecturer:

dr. Caspar J. Van Lissa

Lab sessions

Ward Eiling

Contact hours

This course consists of 8 lectures (HC) and 8 lab sessions (WC).

During the lectures, we mostly discuss the literature assigned for that week. Before each lecture, you will complete a small multiple choice exam (4 questions) about that week's reading which contributes to your grade.

During the lab sessions you will work on applied exercises supervised by a teaching assistant. These exercises are always related to topics discussed during the lectures, and most of them contribute to your final assignment - either directly, or by teaching you skills to apply to your own assignment as part of the homework.

Software

The official software for this course is R, which you should install on your laptop as explained in Appendix A.

1 The Theory Crisis

1.1 Weekly Readings

- Muthukrishna, M., & Henrich, J. (2019). A problem in theory. Nature Human Behaviour, 3(3), 221–229. https://doi.org/10.1038/s41562-018-0522-1
- Eronen, M. I., & Bringmann, L. F. (2021). The Theory Crisis in Psychology: How to Move Forward. Perspectives on Psychological Science, 16(4), 779–788. https://doi.org/10.1177/1745691620970586

1.2 Tutorial

1.2.1 Find a Group

Pick a card that matches your study program:

- Psychology
- Sociology
- Organizational Studies
- Global Management of Social Issues
- Human Resource Studies



Pick a card that matches your *preferred or default* role in team work (you don't have to stick to this role):

- Leader / Visionary
- Monitor / Arbitrator
- Facilitator / Notetaker / Timekeeper
- Devil's Advocate



Now, mingle and try to find group mates with *similar* study programs, and complementary roles in team work. Groups should comprise 3-4 people.

1.2.2 Find a topic

With your group, prepare a text document. At the top of the first page, write:

- Tutorial number (Tutorial 1)
- Group name (come up with your own name)
- Your names and student numbers
- Date

The purpose of this assignment is to find a topic of interest that you want to theorize about, identify what you want your eventual theory to explain, and collect some literature.

1.2.2.1 Step 1

Credit: *This tutorial is based on Noah van Dongen's course "Fundamentals: Becoming better at explaining and understanding".*

As a group, choose a topic to theorize about. Choose a topic within your field of research, or one that touches upon your field of research. The topic may be quite general, such as "the psychopathology of depression" or "territorial behavior". It could also be more specific, such as "helping behavior in humans" or "empathy in conflict resolution". The two restrictions are that A) it can be related to something observable, directly or indirectly. Specifically, you should be able to learn something about the topic you have chosen through observations or experiments. B) You should be able to find sufficient literature (e.g., research) on the topic (see the Fourth part of this assignment). Furthermore, it is recommended to have some experience or knowledge about the topic you choose. Note that you don't have to be a psychology student to choose (some aspect of) psychology, but it makes things easier if you are at least familiar with some of the research that has been done on the topic. Also, it helps if you choose a topic you find interesting and don't mind thinking about. Write down the topic you have chosen and explain in a few sentences (max 100 words) what the topic is about.

1.2.2.2 Step 2

Think of three **phenomena** (Nak et al., 2026) related to your topic you would like to explain. Describe these phenomena in a few sentences. List the phenomena's components/characteristics. Indicate on what the material is based (e.g., a published empirical study, Wikipedia, personal experience, intuition). Note that you do not need to explain these phenomena yet. These phenomena should be as independent of theory (any theory) as possible. Scientists may disagree on how to explain a phenomenon, and even on its ontological status (e.g., whether it reflects a natural kind or a social construction). Still, they should be able to agree on the empirical pattern the phenomenon describes.

1.2.2.3 Step 3

Do a basic literature search on your chosen topic. Make sure that you have both general sources (e.g., textbook chapters and encyclopedia entries, such as Wikipedia) and specific sources (e.g., scientific journal articles). The aim is to have five highly relevant sources. If you cannot find five sources, check with your groupmates and the tutorial teacher and consider picking a different topic.

List the sources you have found and add hyperlinks or other reference information. For each source, make a note of why you think it is relevant. A few words or a sentence is enough. You do not have to read the whole articles yet.

Having a good search strategy is important. Indicate which search engines and/or databases you used (e.g, Google, Google Scholar, Web of Science) and the (combinations of) search terms you used.

1.2.3 Example

Tutorial 1
Group: The Tacky Theoreticians
Names: Caspar van Lissa (123456), Noah van Dongen (789012)
Date: 02-08-2026

1.2.3.1 Step 1

Social behavior of humans. Homo sapiens are social animals that respond to one another and their behavior is shaped by others that are in their direct vicinity. People also respond differently to situations when they are alone versus when there are other people present. The development of a person's (cognitive) behavior is also in part determined by the presence and behavior of other people. I think my main focus will be on helping behavior.

1.2.3.2 Step 2

A. Bystander effect. The probability that somebody helps a person in need decreases with the number of other people present (Blair, Thomson, & Wuensch, 2005). The individual components/characteristics are:

    1. Without other people present, the tendency (probability or frequency) that a person helps a victim is high.
    2. The tendency to help a victim decreases monotonically with the number of other people (bystanders) present.
    3. The tendency to help a victim never drops to 0.

B. Bystander effect when the victim is in danger. The bystander effect varies across different types of situations. The bystander effect is attenuated when the situation is perceived as dangerous for the victim (Fischer et al., 2011). The individual components/characteristics are:

    1. Without other people present, the tendency (probability or frequency) that a person helps a victim is high.
    2. The tendency to help a victim decreases monotonically with the number of other people (bystanders) present.
    3. The tendency to help a victim never drops to 0.
    4. The size of the bystander effect is decreased when the situation is perceived as dangerous to the person in need of help.

C. Bystander effect and danger to the helper. The probability of helping a victim declines as the dangerousness of helping increases (e.g., violent crime with a weapon), in comparison to situations that are less dangerous for the person offering help (e.g., sexual assault with no weapon) (Hart & Miethe, 2008). The individual components/characteristics are:

1. Without other people present, the tendency (probability or frequency) that a person helps a victim is high.
2. The tendency to help a victim decreases monotonically with the number of other people (bystanders) present.
3. The tendency to help a victim never drops to 0.
4. The size of the bystander effect is decreased when the situation is perceived as dangerous to the person in need of help.
5. The size of the bystander effect varies with situational context. The bystander effect is smaller for sexual assault without a weapon than for violent crime with a weapon.

1.2.3.3 Step 3

Wikipedia page: https://en.wikipedia.org/wiki/Bystander_effect

a. An encyclopedia page that contains general information and lists additional references Britannica page: https://www.britannica.com/topic/bystander-effect/Diffusion-of-responsibility
b. An encyclopedia page that contains general information Fischer, P., Krueger, J. I., Greitemeyer, T., Vogrincic, C., Kastenmüller, A., Frey, D., Heene, M., Wicher, M., & Kainbacher, M. (2011). The bystander-effect: A meta-analytic review on bystander intervention in dangerous and non-dangerous emergencies. Psychological Bulletin, 137(4), 517–537. https://doi.org/10.1037/a0023304
c. Is a meta-analysis of studies on the bystander effect and thus contains many references to studies on the bystander effect.

We used Google, Google Scholar, and Web of Science to search for relevant literature. We first used the search term "bystander effect" on Google Scholar, which already produced many hits (39100). We looked at the first three pages and selected those with titles that indicated they were about the bystander effect in (social) psychology. The expectation was that literature on the bystander effect would also be related to social behavior in a more general sense. We then searched on Web of Science to see if any further interesting titles appeared on the first three pages that we missed in Google Scholar. Finally, we searched with Google using the same search term to find more general non-scientific literature that could be of interest, again looking at titles on the first three pages. We think this was an adequate strategy, because we were only searching for a few information-dense and general items and not an exhaustive literature search. We expected that these would show up on the first few pages of the searches, because they most likely would have the term "bystander effect" in the title (Google, Google Scholar, Web of Science) and many places throughout the text (Google). Based on the name of the journal, we selected those that were about (social) psychology.

References

- Blair, C. A., Thompson, L. F., & Wuensch, K. L. (2005). Electronic Helping Behavior: The Virtual Presence of Others Makes a Difference. Basic and Applied Social Psychology, 27(2), 171–178. https://doi.org/10.1207/s15324834basp2702_8
- Fischer, P., Krueger, J. I., Greitemeyer, T., Vogrincic, C., Kastenmüller, A., Frey, D., Heene, M., Wicher, M., &
- Kainbacher, M. (2011). The bystander-effect: A meta-analytic review on bystander intervention in dangerous and non-dangerous emergencies. Psychological Bulletin, 137(4), 517–537. https://doi.org/10.1037/a0023304
- Hart, T. C., & Miethe, T. D. (2008). Exploring bystander presence and intervention in nonfatal violent victimization: When does helping really help? Violence and Victims, 23(5), 637–651. https://doi.org/10.1891/0886-6708.23.5.637

1.3 Homework

This week, you will select one "canonical version" of an existing theory, or draft your own theory from scratch (optional).

> Roll Your Own Theory
>
> The assignment assumes that you will select an existing theory to improve upon. If you draft your own instead, go through all the steps *as if* you are starting from an existing theory - but instead of selecting sources from a published theoretical paper, you write your own. You may find it helpful to:
>
> - Select relevant quotes from *empirical papers* instead, to substantiate the assumptions of your theory.
> - Compare your work to that of other groups who have started with an existing theory.
> - Come back and revise the work you've done this week, once you have gained a better understanding of the next steps in coming weeks.

Credit: *This assignment is based on the Proposition Based Theory Specification instructions (Glöckner et al., 2025).*

Read pages 1-4 of the Proposition Based Theory Specification (PBTS) Guideline.

1.3.1 PBTS Step 1

With your group, select source material for the theory (and specific version of it) to be specified.

a. Decide which theory, and which version of the theory (e.g., original, revised, or latest) will be specified, including the name of the theory, the author, and the original reference. For example, I used Self-Determination Theory, a leading theory of motivation in social psychology.
b. Select one canonical source for the theory specification (e.g., the original article or book chapter). If the theory is not fully specified in one source, or if you are drafting your own theory from scratch, select 2-3 key sources instead, such as relevant journal papers, or textbook chapters. For example, I used Ryan (2023).
c. If the original theory is highly extensive (e.g., published as a book), simplify the task by focusing on its core content as represented in typical journal articles or summary chapters.
d. Select specific lines within the source material that cover the theory's key causal relations, concept definitions, operationalizations, and auxiliary assumptions. Create a spreadsheet like this with these selected sentences, label them (e.g., S1) and document the paper, page number, and ideally line number. The final version of Table 1 should include all agreed sources with page/line information, DOI, and, for books, the edition used. For example, see this spreadsheet
e. All raters should agree on the selected sources. In cases of uncertainty, debate whether the sentence is indeed relevant to the theory.
f. Write 100 words on your approach to Source Selection for your final paper.

2 What Makes a Good Theory?

2.1 Weekly Readings

- Meehl, P. E. (1990). Appraising and Amending Theories: The Strategy of Lakatosian Defense and Two Principles that Warrant It. Psychological Inquiry, 1(2), 108–141. https://doi.org/10.1207/s15327965pli0102_1
- Guest, O. (2024). What Makes a Good Theory, and How Do We Make a Theory Good? Computational Brain & Behavior. https://doi.org/10.1007/s42113-023-00193-2

2.2 Tutorial

2.2.1 Analyze Theory Using Meehl's Taxonomy

In this exercise, you will evaluate the sentences you selected from a published theory using Meehl's hierarchy of progressively stronger theory specifications. The goal is to further your understanding of the function and level of specificity of each sentence. Meehl's orginal taxonomy is:

I. Type of entity postulated
II. Compositional, developmental, or efficient-causal connections among entities
III. Signs of first and second derivatives of functional dynamic laws
IV. Ordering relationships among the derivatives
V. Signs of mixed second-order partial derivatives (interactions)
VI. Functional forms (e.g., linear, logarithmic, exponential)
VII. Trans-situationality of parameters
VIII. Quantitative relations among parameters
IX. Numerical values of parameters

To simplify the assignment, we simplify Meehl's taxonomy as Van Lissa et al. (2026) did:

I. statements about the types of entities postulated (i.e., ontology)
II. statements about causal connections between those entities
III. statements about the functional form of those connections (Levels 3–8)
IV. statements about their specific numerical values (Level 9)

2.2.1.1 Part 1. Classify the theory statements (15 minutes)

First individually, for each selected sentence:

- Read the sentence carefully.
- Decide which of the levels it best represents.

2.2.1.2 Part 2. Evaluate the theory (10 minutes)

As a group:

- Compare your classifications
- Which levels of Meehl's hierarchy are well represented?
- Which levels are missing?
- How strong do you think the theory is according to Meehl's framework?

2.2.1.3 Part 3. Compare theories (15 minutes)

Each group gives one example of every level they identified in their source material, as well as an example of a sentence you were not able to classify (if applicable). Discuss with the class.

2.2.2 Is This Theory "Good"?

Apply Olivia Guest's four-category ontology to critically evaluate a published psychological or social science theory.

2.2.2.1 Part 1. Evaluate Theory along Each Category (10 minutes)

Each group member picks one unique category, and independently answers the guiding questions for that category in Table 1.

Decide whether the theory is "good", adequate, or "bad" in your category.

2.2.2.2 Part 2. Discuss and Prioritize (15 minutes)

Discuss your findings with your groupmates. Make a score card for the theory, rating it as good, adequate, or bad on the four categories.

Then, prioritize the four categories. How important do you think it is that the theory be "good" on each of the four categories? Use this to guide your further work on the theory during coming weeks.

2.2.2.3 Part 3. Class discussion (15 minutes)

As a class, discuss:

- If you had to choose between a theory that scores highly on Meehl's hierarchy but poorly on Guest's ontology, and one that does the opposite, which would you consider the better scientific theory? Why?
- You can pick one criterion out of Meehl's hierarchy and Guest's ontology. Which one do you pick, and why?

2.3 Homework

This week, you will extract IF-THEN statements from the sources you selected last week.

Credit: *This assignment is based on the Proposition Based Theory Specification instructions (Glöckner et al., 2025).*

2.3.1 PBTS Step 2

Individually, fill out Table 2. Translate each source sentence into a set of implications (i.e., IF-THEN statements) and all included propositions that together fully describe the theory.

a. Propositions are statements concerning measurable and non-measurable concepts that can be true or false, and represent an implication's antecedence and consequence. E.g., "A person's needs are satisfied" is a proposition.
b. Propositions can be combined by any logical or mathematical operators, qualifiers, or, if necessary, user-defined uncommon operators (that have to be defined and operationalized).
c. It is often possible to make certain conditions of a theory either part of the propositions or the definition/operationalizations. To keep the propositions simple, these conditions should be part of the definition/operationalizations if they do not belong to the core causal relations of the theory.
d. In this step, remain as close as possible to the original verbal narrative while interpreting the text in accordance with the authors' intended meaning.

(f) In the "Reference" column, reference the label of the text snippet from last week's Table, e.g. S1, S2 ... Sn.

g. If the original specification mentions the functional form of a relationship, put this information in the Comment column.

2.3.2 Combining Specifications-BINGO

As a group, read each other's specifications. Select one student's work to be the "starting point" for the definitive version. The author of this version reads each of their implications aloud. Other students put a check mark in front of their implications if these match the implication read by the first student. If they are clearly similar but there are important differences - discuss these and find agreement on which version of the implication to keep in the definitive version.

When the first student is done reading all of their implications, the other students may have some remaining un-checked implications. That is to say: Implications that they coded, but the first student did not. Discuss these to check whether they should be included in the definitive version or not.

The end product should be a complete table of all unique implications that all students (more or less) agree upon.

If there are important disagreements - keep notes, because this is good material for the Discussion section of your final report.

> Roll Your Own Theory
>
> If you are making your own theory, do the assignment as described above - but instead of coding selected sentences, each student lists all implications that *they think should be part of the theory*. The group activity then constitutes of discussing these and finding agreement about a set of implications.

3 Strategic Ambiguity

3.1 Weekly Readings

- Frankenhuis, W. E., Panchanathan, K., & Smaldino, P. E. (2023). Strategic Ambiguity in the Social Sciences. Social Psychological Bulletin, 18, 1–25.
- Healy, K. (2017). Fuck nuance. Sociological Theory, 35(2), 118-127.

3.2 Tutorial

3.2.1 Ambiguity Battle (30 min, 2x 15)

Collaborate with 4 students, divided into two duos: Theorists VS Challengers. One duo chooses one of the following phenomena and writes down a one-sentence theory to explain the phenomenon:

- Venus effect: most people may hold beliefs that are inconsistent with observable phenomena
- Social facilitation: people do better on tasks when they are with other people rather than when they are doing the task alone.
- The frequency illusion: people notice a specific concept, word, or product more frequently after recently becoming aware of it.
- Barnum effect: individuals give high accuracy ratings to descriptions of their personality that supposedly are tailored specifically to them, yet which are in fact vague and general enough to apply to a broad range of people. This effect can provide a partial explanation for the widespread acceptance of astrology, fortune telling, aura reading, and the Myers-Briggs "personality test".
- The bystander effect: individuals are less likely to offer help to a victim in the presence of other people.
- Dunning–Kruger effect: people with low ability in a specific area systematically give overly positive assessments of their ability, while high performers to underestimate their skills.
- The Group Project Paradox: Everyone in the group feels that they did disproportionally more work than the others.

The other duo brainstorms on "challenges" to the theory, for example:

- Can it explain a null result?
- Can your theory explain cross-cultural variation?
- Can it explain developmental change?
- Can it explain contradictory results in experimental versus observational/naturalistic settings?

You may also describe a specific challenge, for example "My study found that people with high agreeableness do not display the Dunning-Kruger effect".

After each challenge, the first group may revise their theory - but without violating the constraint that it must be a single sentence.

The theorists win if the challengers cannot come up with new challenges. The challengers win if the theorists cannot ammend their theory while respecting the one-sentence constraint.

Switch sides after 10 minutes.

3.2.2 Abstraction VS Nuance (30 min)

Starting with the one-sentence theories from the previous exercise, go around in a circle.

1. In turn, each student mentions their personal experience in relation to the theory (so: 3 or 4 rounds per theory).
2. The other students vote on whether this is consistent with the theory, or whether the theory should be made "more nuanced" to accommodate it.
3. All together, amend the theory if necessary.

Do this for 10 minutes, so for 2-3 theories. Then discuss:

- When did each theory stop becoming better?
- Did any additions genuinely improved explanation?
- Which merely increased complexity?

3.2.3 Ambiguous Causal Claims (30 min)

Causal claims are an important part of (social scientific) theories. However, there are strong norms against explicit discussion of causality in the social science literature. The result is that causal language is often ambiguous. Norouzi et al. (2025) showed that it is possible to automatically extract causal sentences from published papers despite this ambiguity.

The sentences below are examples from the research by Norouzi and colleagues.

With your group, decide for each sentence and keep notes on:

a. Whether it is Causal (C) or Non-causal (NC)
b. Whether it is Ambiguous (Am) or Straightforward (S)
c. Whether it is Nuanced (N) or Abstract (Ab)

- We find differences in the behavior of our experimental participants across the three countries, and we find that Hofstede's dimensions help to explain these differences.
- In the standard symmetric independent private values model in which $_i(d_i) = 0$, Myerson's (1981) revenue equivalence result shows that, given that the charity never keeps the prize, it maximizes its revenue if it always allocates the prize to the donor with the highest marginal revenue.
- First, models of information signaling have shown that sequential giving can have beneficial effects for voluntary public good provision when the common value of the public good is uncertain.
- Even though the level of regulation a facility faces does not depend on the overall size of a facility, the two factors are highly correlated.
- For example, in social dilemmas with random noise, people lack the ability to judge with certainty if others intended to cooperate or defect (Klapwijk & Van Lange, 2009;Van Lange, Ouwerkerk, & Tazelaar, 2002).
- The presence of a threshold constraining participants' activities will lead participants to be less holistic in their cognitive orientations than when a threshold is not present.
- This result is consistent with a previous study which suggests that behaving dishonestly leads to the forgetting of rules.
- Violent media may cause viewers to perceive greater anger and less gratitude during a cooperative task and as such be less inclined to cooperate.

Solution

- We find differences in the behavior of our experimental participants across the three countries, and we find that Hofstede's dimensions help to explain these differences. NC, S, N
- In the standard symmetric independent private values model in which $_i(d_i) = 0$, Myerson's (1981) revenue equivalence result shows that, given that the charity never keeps the prize, it maximizes its revenue if it always allocates the prize to the donor with the highest marginal revenue. C, S, N
- First, models of information signaling have shown that sequential giving can have beneficial effects for voluntary public good provision when the common value of the public good is uncertain. C, Am, N
- Even though the level of regulation a facility faces does not depend on the overall size of a facility, the two factors are highly correlated. NC, S, Ab
- For example, in social dilemmas with random noise, people lack the ability to judge with certainty if others intended to cooperate or defect (Klapwijk & Van Lange, 2009;Van Lange, Ouwerkerk, & Tazelaar, 2002). NC, S, N
- The presence of a threshold constraining participants' activities will lead participants to be less holistic in their cognitive orientations than when a threshold is not present. C, Am, N
- This result is consistent with a previous study which suggests that behaving dishonestly leads to the forgetting of rules. C, S, Ab
- Violent media may cause viewers to perceive greater anger and less gratitude during a cooperative task and as such be less inclined to cooperate. C, S, Ab

Discuss:

- How would you go about writing about causality in your work, after having seen these examples?
- What do you think is the difference between ambiguity and nuance?

3.3 Homework

This week, you are going to translate your IF-THEN specifications from previous weeks into a graph—the technical name is a **Directed Acyclic Graph (DAG)**. In this graph, you connect concepts (propositions, in the language of PBTS) with one-sided arrows to indicate the direction of causality. We will return to the DAG in Week 5 and Week 7.

Note that IF-THEN connections imply causal connections, so the statement "IF a person's need for autonomy is satisfied THEN they experience well-being" implies a causal link of the form `{autonomy satisfied} -> {well-being}`.

3.3.1 Example: Self-Determination Theory

Let's examine a snippet from self-determination theory, and code it for implied causality. The causes are coded in green, and outcomes in blue:

> "For these natural, active **processes of intrinsic motivation and integration** to operate effectively toward **healthy development and psychological well-being**, human beings need particular **nutriments** – both biological and psychological (Ryan, 1995). In the relative absence of such **nutriments**, these natural **processes** will be impaired, resulting in **experiences, development, and behaviors** that are less than optimal" (Lange et al., 2012, p. 417)

The text is a bit ambiguous, and possibly redundant. We see that **processes**, in the first sentence, refers to intrinsic motivation and integration. This invites the question of the relationship between intrinsic motivation and integration - do they go hand in hand (i.e., just two examples of processes set in motion by nutriments)? Are they distinct but similarly related to other constructs? It's not clear. Furthermore, in the second sentence, we find the word **processes** again, but this time without explicit reference to intrinsic motivation and integration. For now, we will assume that intrinsic motivation and integration are distinct but have similar relationships to other constructs, and that the word is used consistently across both sentences.

Nutriments is defined elsewhere in the text - it appears to refer to refer to the three basic needs, as well as biological necessities.

Experiences, development, and behaviors is not well-defined, but we might assume it refers back to **healthy development and psychological well-being**.

In Week 2, we learned to code snippets like this for IF-THEN connections according to PBTS. This would give the following codings:

N	IF	THEN
1	nutriments are absent	intrinsic motivation AND integration are impaire
2	[intrinsic motivation OR integration] is present	healthy development AND psychological well-bei
3	needs are satisfied or thwarted	psychological well-being of all people is affected
4	rewards given	intrinsic motivation can decrease
5	external event is expected to thwart the basic needs	external perceived locus of causality
6	external perceived locus of causality	undermined intrinsic motivation
7	external event is expected to support the basic needs	internal perceived locus of causality
8	internal perceived locus of causality	enhanced intrinsic motivation

We can translate these IF-THEN statements into causal links between specific constructs.

This translation reveals some ambiguities. For example, each IF-THEN relationship is often repeated twice; once phrased in the positive and once in the negative. Many propositions are

repeated, once phrased in a positive way, and once phrased in a negative way. For example, *human beings [need] nutriments [. . .] for these processes to operate* positively phrases a link from nutriments to processes, whereas *In the [absence of] nutriments [. . .] processes will be impaired* negatively phrases the same link. This can be confusing, especially if the two phrasings differ slightly. Here, it seems safe to assume that both sentences have the same meaning.

Another ambiguity is the lack of clear and consistent terminology. For example, in proposition #4, we see "rewards" are related to intrinsic motivation. However, in the remainder of the text, rewards are mentioned as just one example of "external events". We assumed that "rewards" is used as an example of the broader class of external events (pars pro toto).

Another ambiguity: we read that "If [an external event] were expected to thwart [basic needs], it [prompts] an external perceived locus of causality". Since the theory otherwise does not explain how people might "expect" that external events will affect them, and this seems to be more of a question for cognitive than social psychology - we made a simplifying assumption: external events affect needs, and also perceived locus of causality.

We applied two transformations:

1. Replace the original text with concept labels, e.g. "nutriments are present" and "needs are satisfied or thwarted" are both replaced with "needs".
2. After this simplification, remove redundant links

Table 3.1: Unique causal connections

from	to
needs	intrinsic_motivation
needs	integration
intrinsic_motivation	healthy_development
intrinsic_motivation	wellbeing
integration	healthy_development
integration	wellbeing
needs	wellbeing
external_event	intrinsic_motivation
external_event	needs
external_event	locus_of_causality
locus_of_causality	intrinsic_motivation

We translated this table to a DAG:

We can plot the DAG as follows:

```
plot(SDT)
```

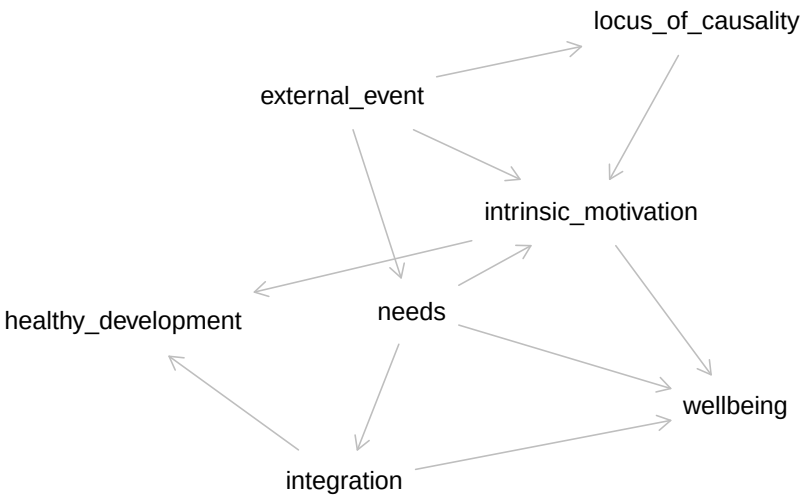


Figure 3.1: Causal diagram implied by SDT

3.3.2 Do It Yourself

Following the example above, you will perform three tasks:

- ☐ Grouping plain-language propositions into a limited set of concepts
- ☐ Revising your `Step_2.csv` table so that it only uses that limited set of concepts
- ☐ Constructing a DAG in R using your revised table.

3.3.2.1 Group Propositions into Concepts

Make a copy of your `Step_2.csv` table, and call it `DAG.csv`.

With your groupmates, discuss:

- Which of the propositions (i.e., the text in the IF- and THEN-columns) are really the same? E.g.: Are "autonomy satisfaction" and "people feel more independent" really the same?
- Are any of them really the same, but positively vs negatively phrased (e.g., "autonomy satisfaction" and "frustration of the need for autonomy")?

- How would you name them if you merged identical propositions? E.g., you could use `autonomy_satisfaction`, or you could go even more abstract and use `need_satisfaction`, in which case you would use the same label for "autonomy satisfaction", "competence satisfaction", and "relatedness satisfaction".

Obviously, these examples are based on SDT; use your own theory for this assignment.

Note: For technical reasons, it is preferable to use short labels without any spaces or special characters. So use `need_satisfaction` , not `satisfaction (of different needs, like autonomy)`.

3.3.2.2 Revise your Table

Now go through your table, `DAG.csv`, and replace every original text with the labels you decided on in the previous step.

3.3.2.3 Make DAG in R

- ☐ Open the folder `patterns_tutorials` you created when following the setup tutorial, Appendix A.
- ☐ Copy your file `DAG.csv` into this folder.
- ☐ Load the RStudio project file, `patterns_tutorials.Rproj`.
- ☐ In the bottom right panel of RStudio, click the `Files` tab, and open the file `make_dag.R`.

Run the following code line by line:

```
library(theorytools)
library(dagitty)
my_theory <- read.csv("DAG.csv", stringsAsFactors = FALSE)[, c(2:3)]
names(my_theory) <- c("from", "to")
# Remove redundant statements
my_theory <- my_theory[!duplicated(my_theory), ]
my_theory <- dagitty(
  paste0("dag {",
  paste0(my_theory$from, " -> ", my_theory$to, collapse = "\n"),
  "}")
)
plot(my_theory)
```

4 Latent Theory in Published Papers

4.1 Weekly Readings

- Norouzi, R., Kleinberg, B., Vermunt, J. K., & Lissa, C. J. van. (2025). Capturing causal claims: A fine-tuned text mining model for extracting causal sentences from social science papers. Research Synthesis Methods, 16(1), 139–156. https://doi.org/10.1017/rsm.2024.13
- Norouzi, R., Kleinberg, B., Vermunt, J., & Van Lissa, C. J. (2026). Can we Automate Social Science Concept Harmonization? A Fine-Grained Review of Performance and Failure Modes (G9pax_v2). PsyArXiv. https://osf.io/preprints/psyarxiv/g9pax_v2/

The lecture sides are here

4.2 Tutorial

The tutorial is here

4.3 Homework

Take the graph you produced during the tutorial, and compare it to the DAG you made in Section 3.3.

Are there any concepts or connections in this new graph that your DAG is missing? Do you want to add these? If so, why (not)? Keep notes on your decision making process for your final paper.

5 Formal Theory Specification

5.1 Weekly Readings

- Smaldino, P. E. (2020). How to Translate a Verbal Theory Into a Formal Model. Social Psychology, 51(4), 207–218. https://doi.org/10.1027/1864-9335/a00042
- Haslbeck, J. M. B., Ryan, O., Robinaugh, D. J., Waldorp, L. J., & Borsboom, D. (2021). Modeling psychopathology: From data models to formal theories. Psychological Methods. https://doi.org/10.1037/met0000303

5.2 Tutorial

In this tutorial, you will go through some of the steps of theory formalization, following the Abductive Formal Theory Construction (AFTC) framework. We will use Self-Determination Theory (SDT) as a guiding example.

The code chunks in this tutorial are also available in the `patterns_principles_exercises` project, in the file `tutorial_formal_theory.R`.

5.2.1 Stage 1: Generating Theory

5.2.1.1 Establish the Phenomena

Below are two excerpts from Ryan (2023):

> "Deci's early studies (e.g., 1971, 1972) were provocative in showing that contingent extrinsic rewards could have negative effects on subsequent interest and persistence (see Ryan, Ryan, & Di Domenico, 2019)."
> "These "undermining" effects of rewards on intrinsic motivation did not occur invariantly but were conditional. As summarized in Deci and Ryan (1980a), rewards have negative effects on intrinsic motivation when used in controlling ways, that is, when applied in order to externally pressure or induce people to perform certain behaviors or meet certain standards. Such controlling reward contingencies can undermine the experience of autonomy essential to intrinsic motivation. In contrast,

rewards can have more positive, or at least no negative, effects on intrinsic motivation when experienced as informational"

Discuss with your group, and describe concisely in your own words, what "phenomena" this excerpt describes.

5.2.1.2 Generate an Initial Verbal Theory

In Week 3, we specified SDT as an Directed Acyclic Graph (DAG). The vignette Formalizing Self-Determination Theory describes how this FAIR theory (See Week 8 for more details) was created based on the original Self-Determination Theory (Deci & Ryan, 2012). A DAG is a diagram that specifies the causal relations proposed by a theory. Such diagrams can be used to derive hypotheses, select control variables for causal inference, and conduct simulation studies (as we will do in this tutorial).

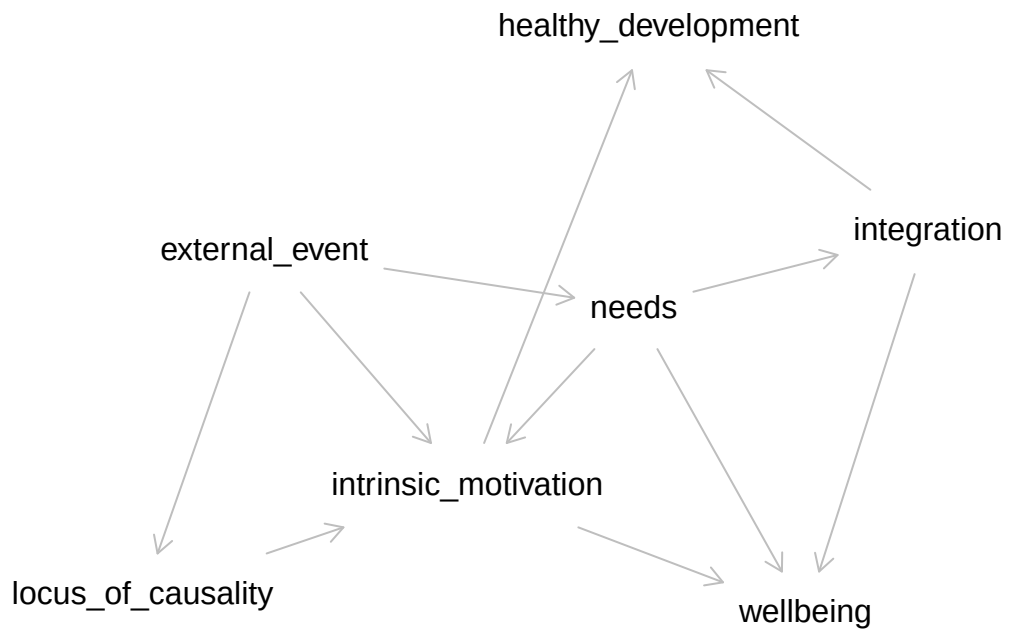
```
library(theorytools)
library(dagitty)
sdt <- dagitty("
dag {
external_event
healthy_development
integration
intrinsic_motivation
locus_of_causality
needs
wellbeing
external_event -> intrinsic_motivation
external_event -> locus_of_causality
external_event -> needs
integration -> healthy_development
integration -> wellbeing
intrinsic_motivation -> healthy_development
intrinsic_motivation -> wellbeing
locus_of_causality -> intrinsic_motivation
needs -> integration
needs -> intrinsic_motivation
needs -> wellbeing
}")
sdt
```

```
dag {
external_event
```

```
healthy_development
integration
intrinsic_motivation
locus_of_causality
needs
wellbeing
external_event -> intrinsic_motivation
external_event -> locus_of_causality
external_event -> needs
integration -> healthy_development
integration -> wellbeing
intrinsic_motivation -> healthy_development
intrinsic_motivation -> wellbeing
locus_of_causality -> intrinsic_motivation
needs -> integration
needs -> intrinsic_motivation
needs -> wellbeing
}
```

We can also plot this DAG:

```
plot(sdt)
```



Discuss with your group:

- Which portions of the DAG are relevant to the phenomenon you've identified?
  - Make a simplified version of the DAG that leaves out all irrelevant components.
- Is the simplified DAG sufficient to explain your identified phenomenon? If it helps, think of the arrows as IF/THEN links. If the causes are present, does the outcome always follow?
  - If not: What additions would you need to make to the DAG to properly describe the phenomenon?

5.2.1.3 Formalize the Initial Theory

We will construct a very basic formalization for the section of SDT that you have selected.

First, we will simplify the DAG. You can do this by hand; suppose we are interested in studying the effect of intrinsic motivation on well-being. We do not need the entire theory to hypothesize about this specific relationship. We can simplify the DAG to derive a restricted version of the theory that includes all variables that might confound the relationship of interest between intrinsic motivation and well-being:

```
sdt_pruned <- dagitty("
dag {
intrinsic_motivation -> wellbeing
external_event -> intrinsic_motivation
needs -> intrinsic_motivation
needs -> wellbeing
}")
sdt_pruned
```

```
dag {
external_event
intrinsic_motivation
needs
wellbeing
external_event -> intrinsic_motivation
intrinsic_motivation -> wellbeing
needs -> intrinsic_motivation
needs -> wellbeing
}
```

Do the same for the phenomenon you identified. When you have done so, we can now create some code to generate a synthetic dataset based on this simplified DAG, using the `simulate_data()` function. This function samples random values for exogenous variables and computes endogenous variables as functions of their predictors (linear functions, by default). By default, regression coefficients are randomly sampled in the range `[-0.6, +0.6]` unless specific values are provided by the user. You might improve the model by choosing more realistic parameter values, for example, using the conventional values for null (0), small (.2), or medium (.4) effect sizes. By default, normal distributions are assumed for exogenous variables and residual error, unless other distributions are specified.

Here, we first generate the code, then examine it in the console, then write it to a file, and finally, run that file to create the simulated data.

```
sim_code <- simulate_data(sdt_pruned, n = 100, run = FALSE)
sim_code
```

```
 [1] "# Set random seed"
 [2] "set.seed(1075769384)"
 [3] "# Set simulation parameters"
 [4] "n <- 100"
 [5] "# Simulate exogenous nodes"
 [6] "external_event <- rnorm(n = n)"
 [7] "needs <- rnorm(n = n)"
 [8] "# Simulate endogenous nodes"
 [9] "intrinsic_motivation <- 0.43 * needs + 0.44 * external_event + rnorm(n = n)"
[10] "wellbeing <- 0.07 * needs + 0.49 * intrinsic_motivation + rnorm(n = n)"
[11] "df <- data.frame("
[12] "external_event = external_event,"
[13] "intrinsic_motivation = intrinsic_motivation,"
[14] "needs = needs,"
[15] "wellbeing = wellbeing"
[16] ")"
```

```
writeLines(sim_code, "formal_model.R")
source("formal_model.R")
```

Put this code into a separate R-file. When you run it all, it will create an object called `df`, which is a table with 100 observations for all variables. Each exogenous variable (causes) is generated by drawing random samples from a normal distribution. Each endogenous variable (effects) is generated by multiplying those exogenous variables with a regression coefficient; larger values mean that the exogenous variable has a larger effect. Then, random noise is added, which represents variability due to other causes not part of your model. If you want this random variability to be smaller, change the `sd` argument of the `rnorm()` function.

5.2.1.4 Evaluating the Initial Theory

Examine the properties of the dataset using any of the following methods, or some of your own choosing:

5.2.1.4.1 Examining descriptive statistics

You can examine the distributions of individual variables using descriptive statistics.

Do not try all of these, instead, discuss which one should capture the pattern you're looking for in the data:

- Univariate distribution, or patterns related to the center and spread of the distribution of one variable: Descriptive statistics and density plots
- Relationships between two variables: Scatter plots and bivariate regression
- Interaction effects (the effect of one variable depends on the value of another variable): Regression with an interaction term (difficult to plot, because that would be a 3-dimensional plot)

5.2.1.4.2 Univariate Distribution

For example, to obtain a summary of all variables in the dataset:

```
desc <- worcs::descriptives(df)
print(desc, digits = 2)
```

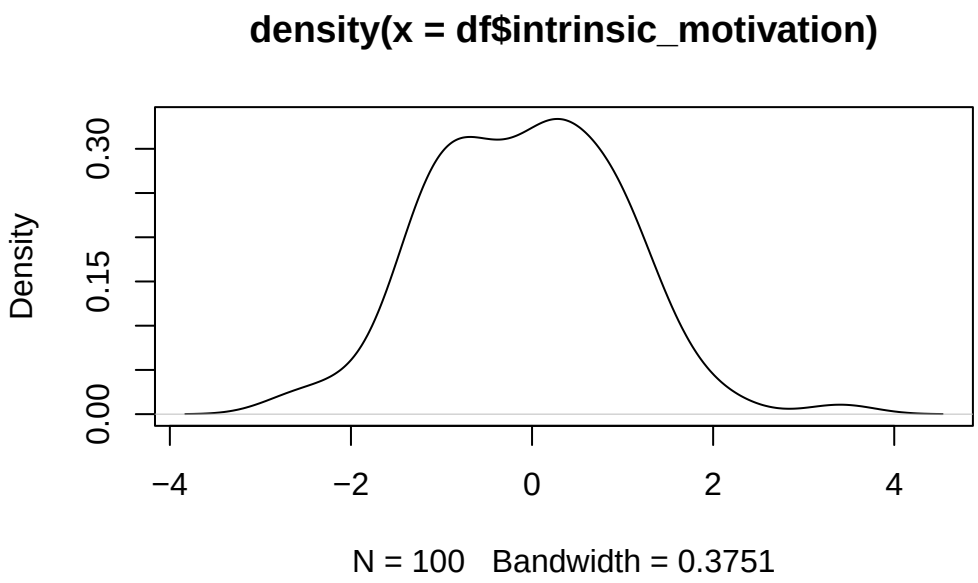
```
                  name    type   n missing unique   mean median   mode
1       external_event numeric 100       0    100  0.104  0.133  0.133
2 intrinsic_motivation numeric 100       0    100 -0.073 -0.011 -0.011
3                needs numeric 100       0    100 -0.069 -0.054 -0.054
4            wellbeing numeric 100       0    100  0.153  0.022  0.022
  mode_value   sd  v  min max range   skew skew_2se  kurt kurt_2se
1         NA 0.90 NA -2.7 2.9   5.6 -0.073    -0.15  0.54     0.57
2         NA 1.05 NA -2.7 3.4   6.1  0.162     0.34  0.25     0.26
3         NA 0.97 NA -2.9 2.5   5.4 -0.062    -0.13  0.53     0.55
4         NA 1.18 NA -2.3 3.5   5.7  0.153     0.32 -0.54    -0.56
```

This gives you information such as the **minimum, maximum, median, mean**.

Use these statistics to look for interesting properties of the data. For example: Does a variable have a particularly high or low mean? Is there a lot of variation between observations? Are some categories much more common than others?

You can also use **base R plots** to visually examine the univariate distribution of an individual variable. For a continuous variable, you can examine its distribution with a density plot:

```
plot(density(df$intrinsic_motivation))
```



Use the plots to look for interesting patterns. For example: Is a distribution symmetric or skewed? Are there possible outliers?

5.2.1.4.3 Relationship between Two Variables

You can test the effect of one variable on another by running a bivariate regression, e.g.:

```
summary(lm(intrinsic_motivation~external_event, df))
```

```
Call:
lm(formula = intrinsic_motivation ~ external_event, data = df)

Residuals:
    Min      1Q  Median      3Q     Max
-2.8071 -0.7514 -0.0675  0.7398  3.3826
```

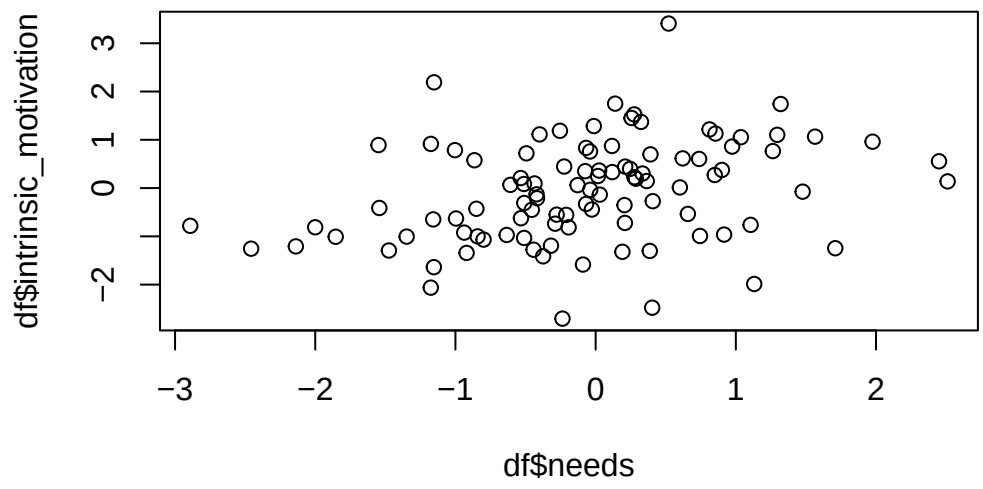
```
Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)     -0.1072     0.1014  -1.057  0.29300
external_event   0.3344     0.1122   2.980  0.00363 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.007 on 98 degrees of freedom
Multiple R-squared:  0.0831,    Adjusted R-squared:  0.07374
F-statistic: 8.882 on 1 and 98 DF,  p-value: 0.003632
```

If the value of `Pr(>|t|)` for `external_event` is smaller than .05, the effect is statistically significant. If the value of `Estimate` is positive, then the effect is positive - if it is negative, then the effect is negative.

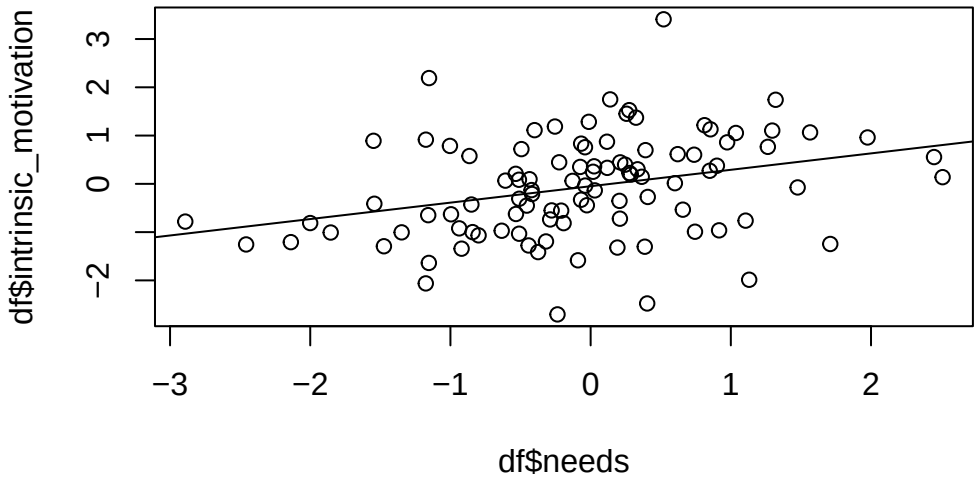
To plot the relationship between two continuous variables, you can use a scatterplot:

```
plot(df$needs, df$intrinsic_motivation)
```



You can add the regression line into this plot to make the direction of the relationship easier to see:

```r
plot(df$needs, df$intrinsic_motivation)
abline(lm(intrinsic_motivation ~ needs, df))
```



5.2.1.4.4 Interaction Effects

You can test whether the effect of one variable on another is contingent on the value of a third variable by running multiple regression regression with an interaction effect, e.g.:

```r
summary(lm(intrinsic_motivation~external_event*needs, df))
```

```
Call:
lm(formula = intrinsic_motivation ~ external_event * needs, data = df)

Residuals:
    Min      1Q  Median      3Q     Max
-2.7558 -0.5784 -0.0377  0.7068  3.1848

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)
(Intercept)          -0.08545    0.09651  -0.885 0.378122
```

```
external_event        0.34018    0.11024   3.086 0.002652 **
needs                 0.35071    0.09925   3.534 0.000632 ***
external_event:needs -0.04237    0.13120  -0.323 0.747459
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9562 on 96 degrees of freedom
Multiple R-squared:  0.1909,    Adjusted R-squared:  0.1656
F-statistic: 7.551 on 3 and 96 DF,  p-value: 0.0001375
```

If the value of `Pr(>|t|)` for `external_event:needs` is smaller than .05, the interaction effect is statistically significant. That means that the effect of `external_event` on `intrinsic_motivation` depends on the value of `needs`.

5.2.2 Stage 2: Developing Theory

The final step is to make changes to your formal theory, to make it better produce the kind of data patterns that your phenomenon implies.

You will probably not have time to do so, but it is possible to make changes to the file `"formal_model.R"`.

For example, you can:

5.2.2.1 Generating a binary variable

You can generate a binary variable using `rbinom()`. For example, if you want to conceptualize `external_event` as the presence or absence of a reward, you could simulate a variable where approximately 50% of observations have a value of 0 and 50% have a value of 1:

```
external_event <- rbinom(n = n, size = 1, prob = 0.5)
```

5.2.2.2 Generating an interaction effect

To make the effect of `external_event` on `intrinsic_motivation` depend on `needs` (e.g., does this external event satisfy needs?), add an interaction term to the equation that predicts `intrinsic_motivation`:

```
intrinsic_motivation <-
  0.56 * external_event +
  0.59 * needs +
  0.80 * external_event * needs +
  rnorm(n = n)
```

The 0.80 specifies the strength of the interaction.

In this example, the effect of external_event on intrinsic_motivation becomes stronger as needs increases.

5.2.2.3 Generating a quadratic effect

To create a nonlinear relationship, include a squared version of a variable:

```
intrinsic_motivation <-
  0.56 * external_event +
  0.59 * needs -
  0.70 * needs^2 +
  rnorm(n = n)
```

The negative coefficient on needs^2 creates an inverted n-shaped relationship: intrinsic motivation initially increases with needs, but at sufficiently high levels of needs it begins to decrease.

A positive quadratic coefficient instead produces a u-shaped relationship.

5.3 Homework

Apply the steps from the tutorial to the DAG you created for your own theory in Section 3.3.

Note that you do not have to apply *all* steps. Think of this week's tutorial as a palette from which you can pick the techniques you need for your specific theory. See how far you can get, and critically reflect on the process.

6 Machine Learning-Informed Theory Construction

6.1 Weekly Readings

- Van Lissa, C. J. (2022). Developmental data science: How machine learning can advance theory formation in Developmental Psychology. Infant and Child Development, 32(6), 1–12. https://doi.org/10.1002/icd.2370
- Rudin, C. (2019). Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. Nature Machine Intelligence, 1(5), Article 5. https://doi.org/10.1038/s42256-019-0048-x

6.2 Tutorial

The topic of this week is to use machine learning to detect patterns in a dataset pertaining to your theory of interest. I have prepared a reproducible example for you based on Self-Determination Theory in the `patterns_principles_exercises` project. The purpose of this tutorial is **not** to give you a foundation in machine learning; that is too complex a topic. However, you should be able to run the code, you are allowed to play around with it and try making some changes, and you might be able to adapt and reuse it for future research projects.

It will probably surprise you that preparing the data for analysis is the most difficult part of this tutorial. Expect to spend most of your time on this step - and **make sure to get the tutorial teacher's approval for your cleaned dataset before you go home**.

6.2.1 Step 1: Find a Dataset (30 minutes)

Search through Google Database Search for a dataset that covers some variables that can be considered operationalizations of the concepts in your theory. One approach is to search the database for the name of your theory. Another approach is to choose one, two, or three construct names from your theory, and search for those.

6.2.1.1 SDT Example: Codebook

I used the search term `"self determination theory"`, and read the descriptions of the datasets until I found the one by Zhao and Xu (2025) on "mental health, academic motivation, and perceived academic performance among Vietnamese students in online learning". It contains operationalizations of a few SDT constructs in the specific context of middle school students' academic stress and physical exercise behavior (we can debate how well this generalizes to SDT in general settings). That's about as good an example as you can expect to find on short notice. My dataset contains the following variables:

Variable	Construct…Description	Type	Coding…Range
ID	Participant identifier; first unnamed column in the spreadsheet	Identifier	1–290
Grade	Grade level	Ordinal categorical	1 = Grade 7 / first-year; 2 = Grade 8 / second-year; 3 = Grade 9 / third-year
Age	Age in years	Numeric	12–15 years
Gender	Gender	Binary categorical	1 = Male; 2 = Female
TS	Teacher pressure, mean score	Ordinal / Likert	Observed 1–7
SOS	Social pressure (social stress), mean score	Ordinal / Likert	Observed 1–7
PS	Parental pressure (stress), mean score	Ordinal / Likert	Observed 1–7
AM	Autonomous motivation, mean score	Ordinal / Likert	Observed 1–7
CM	Controlled motivation, mean score	Ordinal / Likert	Observed 1–7
BI	Exercise intention (behavioral intention), mean score	Ordinal / Likert	Observed 1–7
BH	Exercise behavior, mean score	Numeric / ordinal	Observed values require interpretation from original instrument

6.2.2 Step 2: Preprocess the Data (30 minutes)

You will find a file called `prepare_zhao_2025.R` in the `patterns_principles_exercises` project. Open that file. Each section of code performs a specific data preprocessing step, as

explained in the comments. Work through the file one step at a time to prepare your own dataset for analysis. Note that, for real research projects, you would probably do a lot more data cleaning than this - including solving problems specific to your dataset. For the present tutorial, skip any problem that `prepare_zhao_2025.R` does not handle. Remove every variable that the script cannot process as is, such as variables that cannot easily be recoded into a factor or numeric variable, variables with little or no variation or many missing values, and scales with poor reliability.

6.2.3 Step 3: Run the Analysis (20 minutes)

You will find a file called `analysis_zhao_2025.R` in the `patterns_principles_exercises` project. Open that file, and skim the comments to get a sense of the analysis steps. If you have performed the data cleaning correctly, you should be able to run that file in its entirety without errors, and get machine learning results for your own theory.

6.2.4 Step 4: Interpreting the Results (10 minutes)

We used four machine learning models that capture patterns in the data in different ways. Many other machine learning techniques exist, but most of these are common (except symbolic regression, but it's very interesting nonetheless). In this case, we used:

- **LASSO regression**, which models the outcome as a linear funciton of the predictors (just like normal regression) - but it shrinks the coefficients of less important predictors toward zero. It is easy to interpret because you can read the table of coefficients to evaluate the size and sign of the effect of included predictors, just like a table of regression coefficients. Irrelevant predictors are eliminated from the model.
- A **regression tree**, which captures patterns by repeatedly splitting the data into groups based on predictor values, producing a set of decision rules that can be interpreted directly. Note that, in this tree structure, an interaction effect like X_1 X_2 can be represented by first splitting on one interacting variable X_1, and then on the other, X_2. A nonlinear effect can be represented by splitting on the same variable at different values.
- **Random forests** combine predictions from many regression trees grown on slight permutations of the sample. They also capture potential interactions and nonlinear effects, and often capture much more detail of the patterns in data than a single tree, at the cost of a loss of interpretability. You can explain a forest's predictions using variable importance and partial dependence plots, which describe which predictors matter most for accurate predictions, and how each predictor affects the predictions while averaging across all other predictors, respectively.
- **Symbolic regression** constructs a mathematical expression, given a set of variables and mathematical operators, that reproduces patterns in the data. You can allow it to construct any kind of function, including nonlinear and interacting functions. The final model is an explicit equation, which can be interpreted at face value.

6.2.4.1 Model Fits

The first thing we do is interpret the table with model fits. Here, we use R^2 - which is a bit similar to the R^2 you might know as "explained variance" from linear regression. One key difference is that we have R^2_{train} and R^2_{test}. The R^2_{train} can be straightforwardly interpreted as the percentage of variance explained in the data the model already saw during training. The R^2_{test} is a bit different: it looks at the variance explained by a specific model *in new data* (the test data), relative to just using the mean of the training data to predict new observations.

Model	R^2_{train}	R^2_{test}
LASSO	0.21	0.27
Random Forest	0.86	0.28
Regression Tree	0.68	0.05
Symbolic regression	0.11	0.34

To pick the best model, it's most useful to look at R^2_{test}, because R^2_{train} can be overly optimistic - a sufficiently complex model can literally memorize the correct value for all the training data, achieving a very high $R^2_{train} = 1.00$ but an abysmal R^2_{test}. This phenomenon is called "overfitting".

Quiz

Q1: Which model seems to be most overfit?

A. Random forest
B. Symbolic regression
C. LASSO
D. Regression tree

Q2: Which model is least overfit?

A. Regression tree
B. Random forest
C. Symbolic regression
D. LASSO

Q3: The R^2_{train} is positively biased

☐ True
☐ False

Q4: The R^2_{test} tells you how well your model generalizes to new data.

☐ True
☐ False

Answers

Q1: Regression tree
Q2: Symbolic regression
Q3: TRUE
Q4: TRUE

According to this metric, symbolic regression performs best ($R^2_{test} = 0.34$), followed by the random forest ($R^2_{test} = 0.28$) and LASSO ($R^2_{test} = 0.27$). The random forest is difficult to interpret, although you can analyze its predictions using measures such as variable importance and partial dependence plots. By contrast, symbolic regression and LASSO are straightforward to interpret: symbolic regression yields an explicit equation, and LASSO yields a table of regression coefficients. From a theory construction perspective, these two models are therefore more attractive, because they combine similar or better predictive performance with high interpretability. The regression tree performs poorly ($R^2_{test} = 0.05$): its predictions in new data are only marginally better than using the mean of the test data. We should also be careful not to read too much into the differences between the models, because they are rather small and the ranking may well change in a different sample.

6.2.4.2 Interpret LASSO

To interpret the LASSO model, we look at its regression coefficients. These are interpreted the same as other regression coefficients, except that these are all biased towards zero:

Predictor	b_{1se}
(Intercept)	4.61
Grade	0.00
Age	0.00
GenderF	0.00
TS	0.00
SOS	0.00
PS	0.00
CM	-0.16
BI	0.27
BH	0.00

Quiz

Q1: BI is one of the most influential predictors

☐ True

☐ False

Q2: CM is the least influential predictor

☐ True
☐ False

Q3: PS was dropped from the model

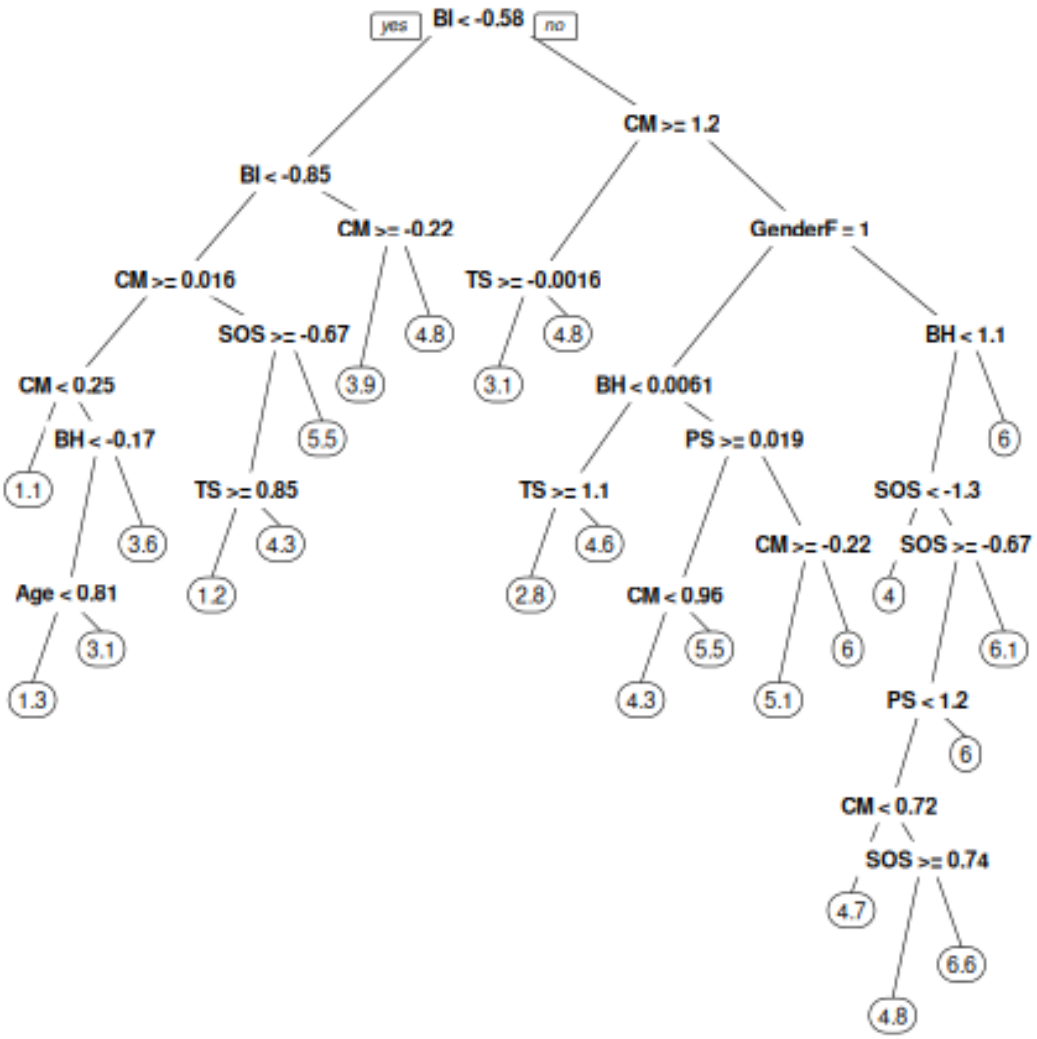
☐ True
☐ False

Answers

Q1: TRUE
Q2: FALSE
Q3: TRUE

6.2.4.3 Interpret a Tree

You can interpret a regression tree by plotting it:



The plot tells us:

- which variables are selected at all; these are the relevant predictors
- how the model comes to a prediction: you navigate the decision tree based on your values on the predictor variables, until you reach the bottom of the tree. The predicted value is the label of the terminal node.

Quiz

Q1: In this model, all variables are used.

☐ True
☐ False

Q2: To get a prediction, you start from the top: if your BI score is not lower than -.58, then go down the right branch

☐ True
☐ False

Q3: The fact that the model first splits on BI and then on CM suggests a potential interaction between these two variables.

☐ True
☐ False

Q4: If the tree had been simpler (= less complex branching), it would be less overfit.

☐ True
☐ False

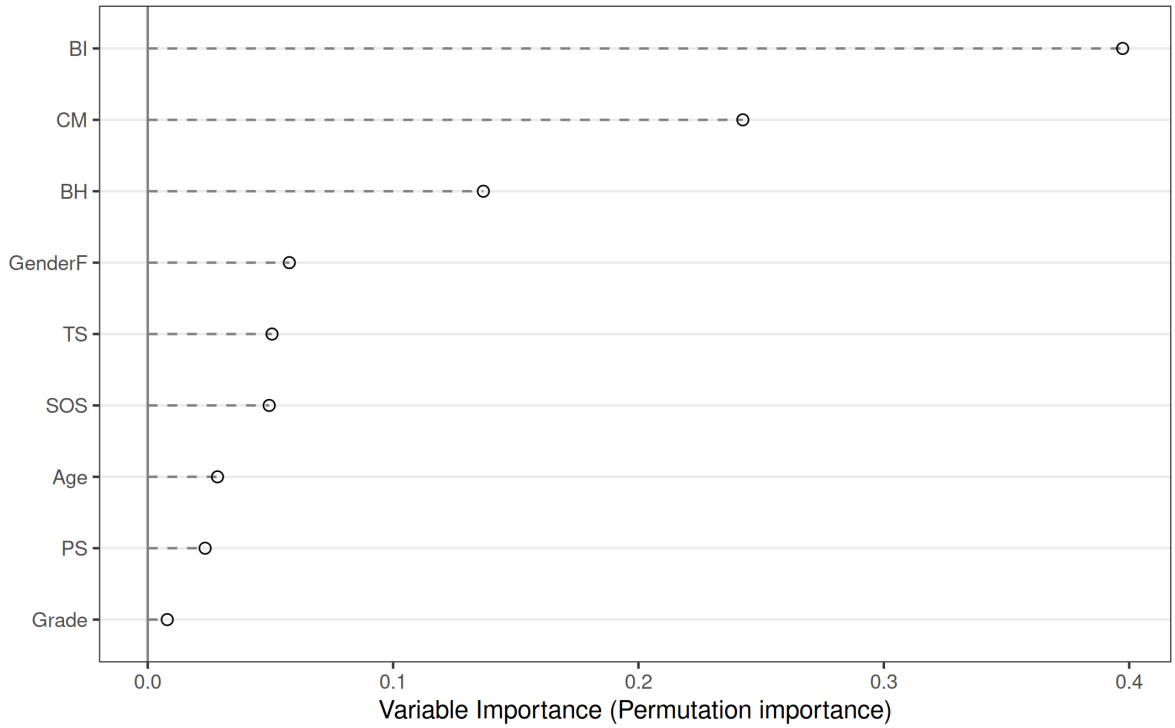
Answers

Q1: FALSE
Q2: TRUE
Q3: TRUE
Q4: TRUE

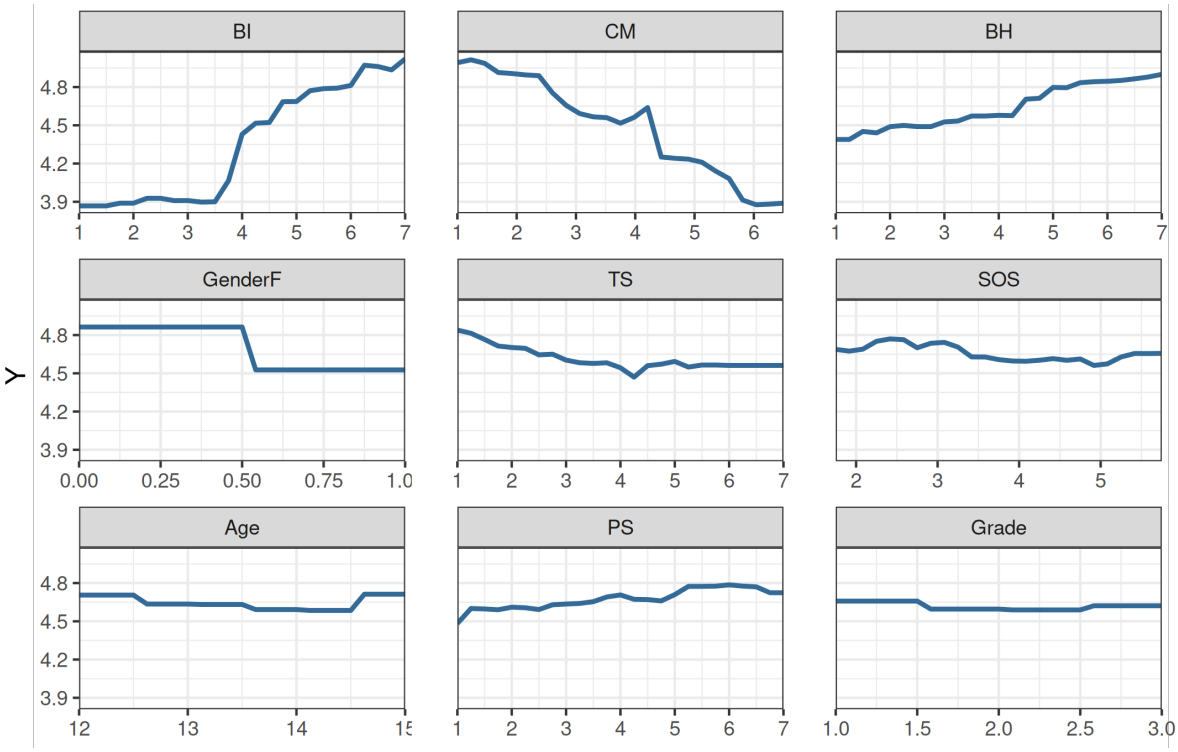
6.2.4.4 Explain a Random Forest

Unlike a single tree, a random forest is not interpretable, but you can analyze/explain its predictions to some extent.

First, you can randomly shuffle each predictor, and see how much *worse* the model predicts when that happens. If shuffling one predictor really diminishes predictive performance, it must have been an important predictor.



To get a sense of the shape of the relationship between each predictor and the outcome (averaging over all other predictors), we can use "partial dependence plots":



Quiz

Q1: Random forests gives a very different picture of what the most important predictors are than LASSO.

- ☐ True
- ☐ False

Q2: Whether you have low or medium levels of BI does not make a difference for your AM (the outcome).

- ☐ True
- ☐ False

Q3: For increasing CM, we see an approximately linear decrease of AM

- ☐ True
- ☐ False

Q4: The effect of GenderF could have any other shape than the step function we see here.

- ☐ True

☐ False

Answers

Q1: FALSE
Q2: TRUE
Q3: TRUE
Q4: FALSE

6.2.4.5 Interpret Symbolic Regression

Finally, interpreting the symbolic regression model is very straightforward; just examine the equation constructed by the algorithm:

BI 4.61

Quiz

Q1: In this case, symbolic regression gives us a linear function

☐ True
☐ False

Q2: The intercept of this function is:

...

Q3: The slope of BI is

...

Answers

Q1: TRUE
Q2: c(4.61, 0.01)
Q3: c(1, 0.01)

6.3 Homework

If you managed to run the analysis, then you should now have machine learning results that tell you something about the predictors of one variable that (loosely) corresponds to a specific concept in your DAG.

Interpret the results you've obtained, and reflect on your previous theory (the DAG). Do you see any predictors that are not in your DAG but should be? Did you learn anything new about the direction of effects, about interaction effects, about the shape of association between concepts in your DAG? Do any of the analyses have weird results that might have an explanation totally unrelated to your theory? Write your reflections down to become part of your final report.

7 Causal Discovery

7.1 Weekly Readings

- Pearl, J. (2019). The seven tools of causal inference, with reflections on machine learning. Communications of the ACM, 62(3), 54–60. https://doi.org/10.1145/3241036
- Vowels, M. J. (2022). A Causal Research Pipeline and Tutorial for Psychologists and Social Scientists (Sections 1-4 and 9). arXiv. https://doi.org/10.48550/arXiv.2206.05175

7.2 Tutorial

Suppose you have a correlation matrix based on observational (i.e., non-experimental) data and nothing else. How much can you say about what causes what?

The standard answer that we learn in statistics courses is "nothing": *correlation does not equal causation*. The surprising answer is "sometimes quite a lot". Certain causal structures leave distinctive traces in the pattern of correlations, and those traces can be read back off the data, granted we are willing to make some assumptions. Methods that do this are called *causal discovery methods*.

In this tutorial, we will explore how to use causal discovery (based on conditional independence testing) to go beyond correlation and infer causal relationships from observational (i.e., non-experimental) data.

```
library(tidySEM)
library(ppcor)
library(qgraph)
library(pcalg)
library(Rgraphviz)
library(dagitty)
library(tpc)
library(webexercises)
```

One helper function

Software implementations of causal discovery methods typically output an adjacancy matrix to represent the causal structure, but they are hard to read. The function below prints a graph as a list of edges instead. Run it once; we use it throughout.

```
edges_of <- function(fit) {
  # coerce the fit into an adjacancy matrix
  m <- as(fit, "amat")
  # store the variable names
  nms <- colnames(m)
  # initialize empty vector to collect edge strings
  out <- character(0)

  # loop over every ordered pair of variable names
  for (i in nms) for (j in nms) {
    # Skips the diagonal (since a variable has no edge to itself)
    if (i == j) next
    # pcalg convention: m[i, j] == 0 and m[j, i] == 1 means i -> j
    # Detects a directed edge
    if (m[j, i] == 1 && m[i, j] == 0) out <- c(out, paste(i, "->", j))
    # Detects a undirected edge
    if (m[i, j] == 1 && m[j, i] == 1 && i < j) out <- c(out, paste(i, "--", j))
  }
  out
}
```

7.3 Part 1: Three patterns, Three Traces (30 min)

When thinking about what traces can be used to infer the causal structure, it helps to distinguish between a few different causal structures containing 3 variables, visualized in the directed acyclic graphs (DAGs; i.e., a causal graph without cycles) below.

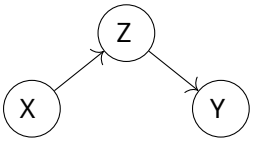


Figure 7.1: Chain

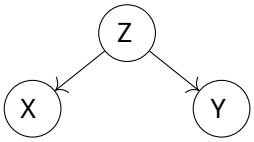


Figure 7.2: Common cause

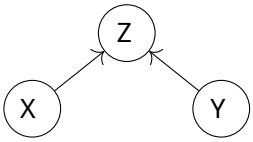


Figure 7.3: Common effect

In this part of the tutorial, we will simulate data for each one, so that we know the true answer

("ground truth") and look at what shows up in the correlations. More specifically, we will take a look at two metrics for each pattern:

- the plain correlation between `X` and `Y`;
- the correlation between `X` and `Y` *after accounting for* `Z`, the **partial correlation**.

> **i** What does "accounting for" mean?
>
> "Accounting for `Z`" means: within groups of people who have the same value of `Z`, are `X` and `Y` still related? The same thing is sometimes expressed using a few different phrasings: (a) "controlling for `Z`", (b) "conditioning on `Z`", or (c) "holding `Z` fixed".

7.3.1 Pattern 1: a chain

In a chain, a third variable `Z` sits between `X` and `Y`. In other words, `Z` is a mediator in the causal relationship between `X` and `Y`. As an example, teacher pressure lowers motivation, and motivation drives behaviour.

```
set.seed(6)
n <- 5000   # large, so that sampling noise does not obscure the pattern
a <- 0.6    # strength of every arrow, on a correlation scale

chain <- local({
  X <- rnorm(n)
  Z <- a * X + rnorm(n, sd = sqrt(1 - a^2))
  Y <- a * Z + rnorm(n, sd = sqrt(1 - a^2))
  data.frame(X = X, Z = Z, Y = Y)
})

cor(chain$X, chain$Y)                                    # related?
```

```
[1] 0.3574731
```

```
pcor.test(chain$X, chain$Y, chain$Z)$estimate          # still related given Z?
```

```
[1] -0.002474125
```

From the output, we can tell that `X` and `Y` are correlated, and that the correlation disappears once we account for `Z`. The intuitive explanation for this is that if the only route from `X` to `Y` runs through `Z`, then holding `Z` fixed blocks (i.e., cuts the flow of association) the route.

7.3.2 Pattern 2: a common cause

With a common cause, a third variable `Z` causes both `X` and `Y`, but there is *no causal relation* between the latter two. In other words, `Z` is a confounder in the relationship between `X` and `Y`.

```
fork <- local({
  Z <- rnorm(n)
  X <- a * Z + rnorm(n, sd = sqrt(1 - a^2))
  Y <- a * Z + rnorm(n, sd = sqrt(1 - a^2))
  data.frame(X = X, Z = Z, Y = Y)
})

cor(fork$X, fork$Y)
```

```
[1] 0.3558507
```

```
pcor.test(fork$X, fork$Y, fork$Z)$estimate
```

```
[1] 0.007863247
```

As an example, we may see that the number of shark attacks and the number of ice cream sales move together (when one tends to be higher, the other tends to be as well), which is all explained by the amount of sunny days. And once we *account* for the amount of sunny days, the correlation between shark attacks and ice cream sales dissappears, as the pathway is blocked.

Note that we practically get the same two numbers as the chain: both produce a correlation that vanishes when you account for the middle variable. This is because *the data cannot tell apart a chain from a common cause.*

7.3.3 Pattern 3: a common effect

When there is a common effect, two variables `X` and `Y` both cause a third variable `Z`, where `Z` is referred to as a *collider*. Note that the arrows are pointed the opposite way as what we saw in pattern 2. As an example, natural talent and working hard both contribute to getting a good exam grade; but the former two do not causally impact one another.

```r
collider <- local({
  X <- rnorm(n)
  Y <- rnorm(n)
  Z <- a * X + a * Y + rnorm(n, sd = sqrt(1 - 2 * a^2))
  data.frame(X = X, Z = Z, Y = Y)
})

cor(collider$X, collider$Y)
```

```
[1] 0.0155854
```

```r
pcor.test(collider$X, collider$Y, collider$Z)$estimate
```

```
[1] -0.5466282
```

Among students who got a good grade, someone with little natural talent must have worked hard, and someone who did not work hard must have had natural talent. So learning one tells you something about the other. In more formal words, when we account/control for exam grades, we create an association between the two previously uncorrelated variables. Here, the initial state is blocked and by accounting for the collider we unblocks/opens up the pathway.

This is exactly the opposite of what we saw for a common cause, where we removed the correlation by accounting for the third variable.

7.3.4 The three traces side by side

```r
trace <- function(d) {
  c(r_XY = cor(d$X, d$Y),
    r_XY_given_Z = pcor.test(d$X, d$Y, d$Z)$estimate)
}

patterns <- data.frame(
  pattern = c("Chain:          X -> Z -> Y",
              "Common cause:  X <- Z -> Y",
              "Common effect: X -> Z <- Y"),
  rbind(trace(chain), trace(fork), trace(collider))
)
print(patterns, digits = 2, row.names = FALSE)
```

```
                    pattern  r_XY r_XY_given_Z
Chain:          X -> Z -> Y 0.357      -0.0025
Common cause:   X <- Z -> Y 0.356       0.0079
Common effect: X -> Z <- Y 0.016      -0.5466
```

The former two patterns are statistically indistinguishable. The third has a different statistical trace: unrelated on their own, but related once we account for the third variable.

This assymetry between the chain and common cause on the one hand and the common effect on the other is the basis of constraint-based causal discovery.

Quiz

Q1: You find that X and Y are correlated, and that the correlation disappears when you account for Z. Which pattern is it?

A. Either one; the data cannot distinguish them
B. A chain
C. A common cause

Q2: You find that X and Y are uncorrelated, but become correlated once you account for Z. Which pattern is it?

A. A common cause
B. A chain
C. A common effect

Q3: Accounting for a common effect can create an association where there was none.

☐ True
☐ False

Answers

Q1: Either one; the data cannot distinguish them
Q2: A common effect
Q3: TRUE

i Accounting for the third variable Z has different consequences

Accounting for the third variable Z blocks a chain (pattern 1) and blocks a common cause (pattern 2), and *unblocks* a collider (pattern 3).

7.4 Part 2: Can the algorithm find a structure we hid? (30 min)

Peter Spirtes and **C**lark Glymour operationalized this assymetry between the three patterns and developed the *PC algorithm*, which tests every pair of variables under every relevant conditioning set, deletes edges whenever it finds independence, then orients arrows using the collider traces from Part 1.

But can algorithmns actually find a structure we hide? And under what conditions? To address these questions, we will hide a structure in simulated data and see whether the algorithm can retrieve this correctly.

We will *simulate* graphs with four variables that were present in the Zhao (2025) data (introduced in Week 6 Machine-Learning Informed Theory Construction; we will come back to the *real* data in Part 3):

- `AM` (autonomous motivation)
- `TS` (teacher pressure)
- `BI` (behavioural intention)
- `BH` (sports behaviour).

```
set.seed(1)
n <- 1000
vars <- c("AM", "TS", "BI", "BH")
alpha <- 0.10 # significance level of independence tests
```

Below we create a simple function for running the PC, algorithm. We can see that the `pc` function only requires the correlation matrix and sample size rather than the raw data (see the `suffStat` line).

```
run_pc <- function(data, alpha = 0.10) {
  pcalg::pc(suffStat = list(C = cor(data), n = nrow(data)),
     indepTest = gaussCItest,
     alpha     = alpha,
     labels    = names(data))
}
```

7.4.1 World 1: a structure containing a common effect

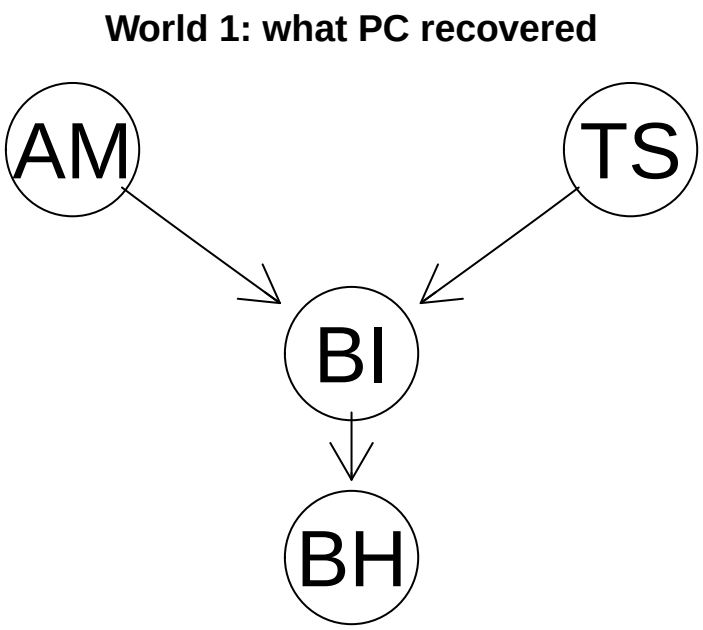
Let us first create a fictional structure that we call World 1, containing a common effect, with the causal relations: `AM -> BI`, `TS -> BI`, `BI -> BH`. Motivation and teacher pressure both cause intention, and intention causes behaviour. Note that `BI` is a common effect of `AM` and `TS`.

```
world1 <- local({
  TS <- rnorm(n)
  AM <- rnorm(n)
  BI <- 0.6 * TS + 0.6 * AM + rnorm(n)
  BH <- 0.6 * BI + rnorm(n)
  data.frame(AM = AM, TS = TS, BI = BI, BH = BH)
})

fit1 <- run_pc(data = world1)
edges_of(fit1)
```

```
[1] "AM -> BI" "TS -> BI" "BI -> BH"
```

```
plot(fit1, main = "World 1: what PC recovered")
```



We can see that PC recovers each arrow correctly. The algorithm found the common effect at `BI` using the trace from Part 1, and once those two arrows were fixed, the direction of `BI -> BH` followed: pointing it the other way would have created a second common effect at `BI`, which the data rule out.

7.4.2 World 2: a pure chain

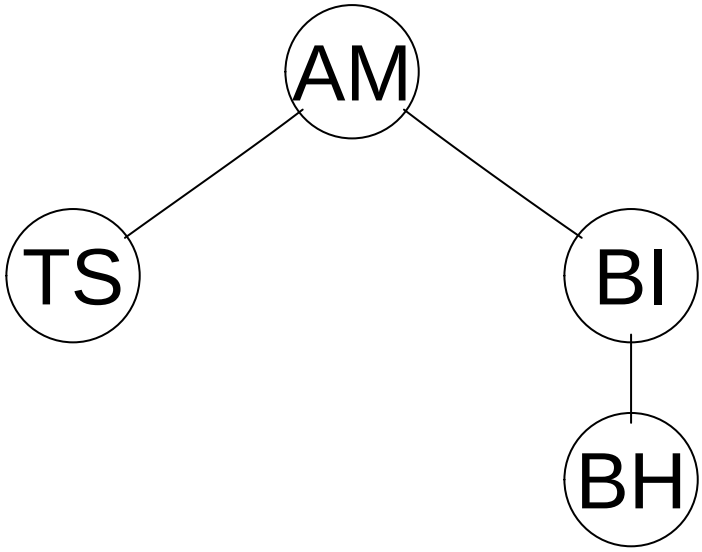
Let's now consider another fictional causal structure, called World 2, with the causal relations: `TS -> AM -> BI -> BH`. These comprise the same four variables, but now with a chain structure rather than a common effect.

```
world2 <- local({
  TS <- rnorm(n)
  AM <- 0.6 * TS + rnorm(n)
  BI <- 0.6 * AM + rnorm(n)
  BH <- 0.6 * BI + rnorm(n)
  data.frame(AM = AM, TS = TS, BI = BI, BH = BH)
})

fit2 <- run_pc(data = world2)
edges_of(fit2)
```

```
[1] "AM -- TS" "AM -- BI" "BH -- BI"
```

```
plot(fit2, main = "World 2: what PC recovered")
```



We can see that PC correctly recovers the skeleton (i.e., undirected graph showing which variables are statistically associated), but cannot tell which way the arrows point. This indeterminacy of the complete causal structure (including orientation) follows directly from what we learnt in Part 1: a chain running left to right and a chain running right to left produce identical correlations, so no amount of data can separate them. Without a common effect, there is no trace to exploit.

7.4.3 How many structures fit equally well?

Let us figure out how many causal structures can explain the correlations that were found in the data simulated based on the structures of World 1 and World 2.

```
n_structures <- function(fit) nrow(pcalg::pdag2allDags(as(fit, "matrix"))$dags)

n_structures(fit1)
```

```
[1] 1
```

```
n_structures(fit2)
```

```
[1] 4
```

We can see that PC can pin it down to a single structure for World 1, whereas it leaves several different possible causal structures for World 2.

Quiz

Q1: PC recovered World 1 exactly because

A. the sample size was large
B. chains are harder to simulate
C. the true structure contained a common effect, which leaves a unique trace

Q2: In World 2, PC left the edges undirected because

A. the algorithm ran out of tests
B. chains pointing in either direction imply the same correlations
C. the associations were too weak

Q3: A graph with undirected edges means the analysis failed.

☐ True
☐ False

Answers

Q1: the true structure contained a common effect, which leaves a unique trace
Q2: chains pointing in either direction imply the same correlations
Q3: FALSE

What the PC provides as output

- The PC algorithmn constructs the skeleton and then orients any collider structures and other compelled edges to construct a DAG. Because we are limited by these collider structures for orientation (we cannot distinguish between a chain and common cause as seen in Part 1; e.g., World 2), the algorithm returns a completed partially directed acyclic graph (**CPDAG**; completed because it contains every single edge orientation that can be deduced)
- A CPDAG is consistent with *multiple candidate DAGs* (where all edges are oriented). These DAGs are statistically indistinguishable. We call this group of DAGs the **Markov equivalence class**. In World 1, the MEC consisted of one structure; in World 2 it consisted of several structures.

7.5 Part 3: Your theory meets real data (30 min)

The PC algorithm uses statistical rules of thumb to construct DAGs; it does not know your underlying theory. At this point you may ask: Do the DAGs constructed by `pcalg::pc()` make sense? If you used only expert knowledge, what DAG would you construct? And does this DAG have the causal implications that allow it to be recovered by algorithms like PC?

We now turn to the *real* Zhao (2025) dataset, using the same four constructs as we simulated data for in Part 2:

- **AM**: Autonomous motivation, as an operationlization of one of the fundamental needs.
- **TS** Teacher pressure (teacher stress), as an operationalization of "external events"
- **BI**: Behavioral intention, which we will interpret as an operationalization of integration.
- **BH**: Sports behavior, as a context-specific operationalization of healthy development.

These are imperfect matches for the theoretical concepts, but for the purpose of a tutorial - that is fine. All four are mean scores aggregrated across questionnaire items, with higher values indicating higher levels of the corresponding construct.

```
data_raw <- readRDS("zhao_2025.rds")
vars <- c("AM", "TS", "BI", "BH")
df <- data_raw[, vars] # select four variables for this tutorial

print(descriptives(df), digits = 2)
```

```
  name    type   n missing unique mean median mode mode_value  sd  v min max
1   AM numeric 290       0     31  4.6    4.6  4.6         NA 1.4 NA   1   7
2   TS numeric 290       0     21  2.9    2.8  2.8         NA 1.1 NA   1   7
3   BI numeric 290       0     16  4.9    5.0  5.0         NA 1.3 NA   1   7
4   BH numeric 290       0     13  4.2    4.5  4.5         NA 1.4 NA   1   7
  range  skew skew_2se kurt kurt_2se
1     6 -0.43    -1.49  3.2      5.6
2     6  0.41     1.43  2.8      4.8
3     6 -0.48    -1.68  3.5      6.2
4     6 -0.22    -0.76  2.8      5.0
```

```
print(cor(df), digits = 2)
```

```
      AM    TS    BI   BH
AM  1.00 -0.25  0.51  0.3
TS -0.25  1.00 -0.31 -0.1
BI  0.51 -0.31  1.00  0.4
BH  0.30 -0.10  0.40  1.0
```

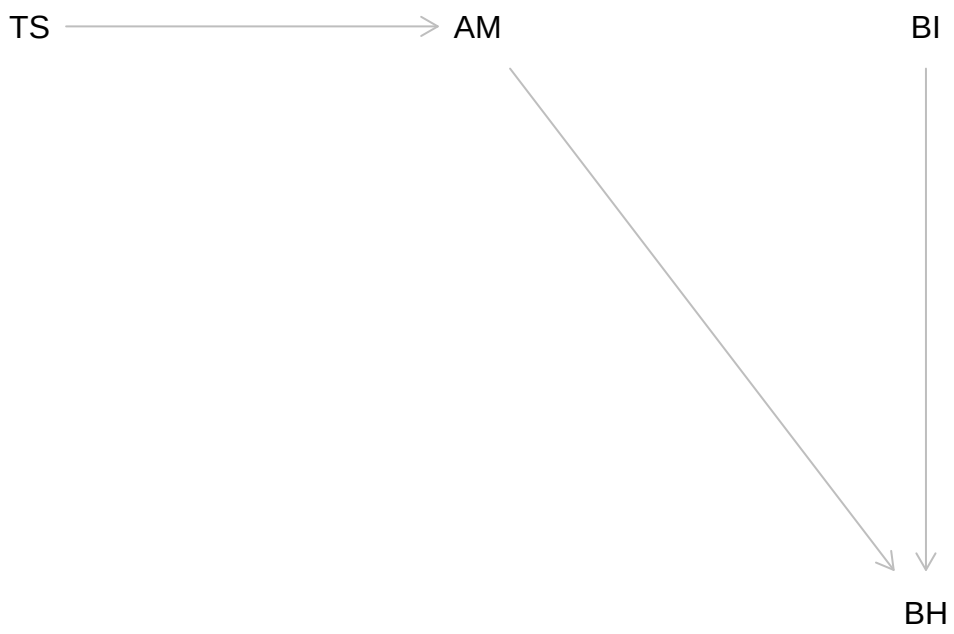
7.5.1 Step 1: Write down your theory

Before looking at any output of the algorithm, we should commit and make our hypothesized/theoretical DAG explicit. Shortly discuss what you think is the DAG is in this substantive setting and edit the code (`theory` line) below accordingly.

```
theory <- dagitty('dag {
  TS -> AM
  AM -> BH
  BI -> BH
}')

coordinates(theory) <- list(
  x = c(TS = 0, AM = 1, BI = 2, BH = 2),
  y = c(TS = 0, AM = 0, BI = 0, BH = 1)
)

plot(theory)
```



7.5.2 Step 2: Ask what your theory predicts

At this point, we want to see what predictions our theory actually makes (i.e., what causal implications it has).

```
dagitty::impliedConditionalIndependencies(theory)
```

```
AM _||_ BI
BH _||_ TS | AM
BI _||_ TS
```

The symbolic notation `X _||_ Y | Z` should be read as "X is unrelated to Y once you account for Z".

Write down the predictions in words.

7.5.3 Step 3: Test the predictions

Let us now test our predictions on the data using the `localTests()` function.

```
round(dagitty::localTests(theory, df, type = "cis"), 3) # cis: linear conditional independenc
```

```
                estimate p.value   2.5%  97.5%
AM _||_ BI         0.505   0.000  0.415  0.592
BH _||_ TS | AM   -0.031   0.604 -0.146  0.085
BI _||_ TS        -0.314   0.000 -0.415 -0.207
```

Each row reports an estimate of the (partial) correlation your theory says should be zero, together with a p-value and a confidence interval. An estimate far from zero, with an interval excluding zero, means the data contradict that particular prediction.

Interpret the result in substantive terms.

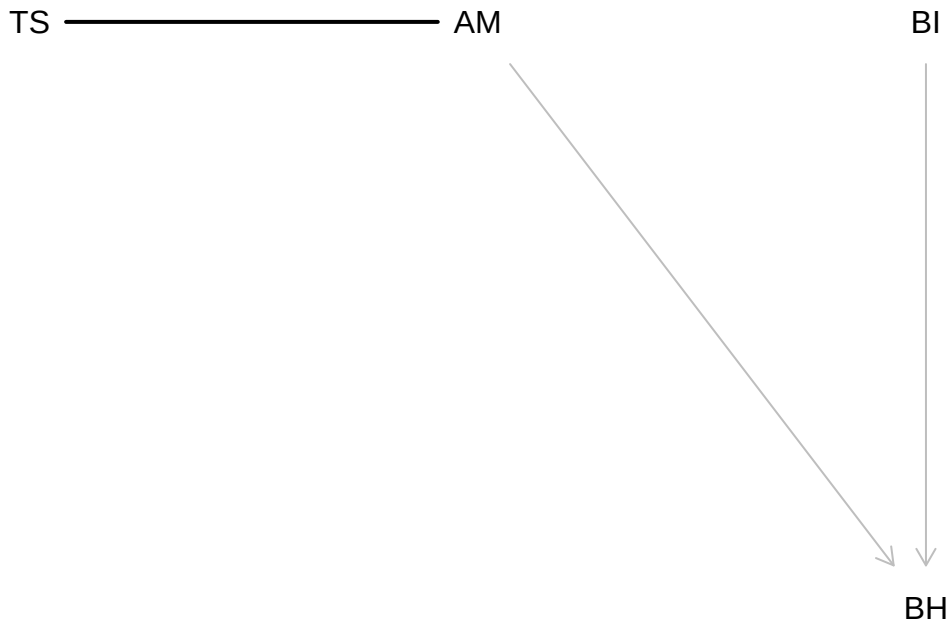
7.5.4 Step 4: What could PC recover if your theory were true?

Suppose your theory is exactly right, all of PC's assumptions hold (see "What PC assumes" below) and the sample size is infinite. Even then, PC can only recover your DAG up to its Markov equivalence class (MEC), as we saw in Part 2. Let's now determine this MEC based on our theoretical DAG.

```
theory_cpdag <- dagitty::equivalenceClass(theory)
theory_cpdag
```

```
pdag {
AM [pos="1.000,0.000"]
BH [pos="2.000,1.000"]
BI [pos="2.000,0.000"]
TS [pos="0.000,0.000"]
AM -- TS
AM -> BH
BI -> BH
}
```

```
plot(theory_cpdag)
```



Edges that remain undirected (`--` in code; dark grey in diagram) are arrows in your theory whose direction PC cannot possibly determine from data. So if PC leaves one of them undirected, we know that this is not evidence against your theory.

7.5.5 Step 5: Let the data propose a structure

Now run PC on the same data and compare.

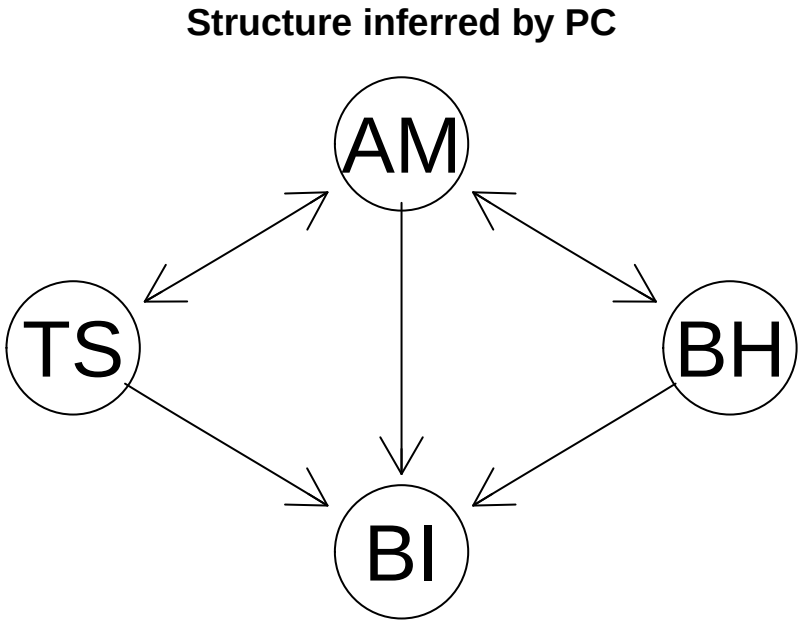
```r
suffStat <- list(C = cor(df), n = nrow(df))

pc_fit <- pc(suffStat  = suffStat,
             indepTest = gaussCItest,
             alpha     = alpha,
             labels    = vars)

edges_of(pc_fit)
```

```
[1] "AM -- TS" "AM -> BI" "AM -- BH" "TS -> BI" "BH -> BI"
```

```r
plot(pc_fit, main = "Structure inferred by PC") # bidirectional arrows imply that the arrow is
```



7.5.6 Step 6: Compare, and account for the differences

Work through these questions in your group and write your answers down.

1. **Which connections do the two graphs agree on?** These are the claims supported both by your theory and by the data.
2. **Which edges does PC include that your theory omits?** Is there a plausible mechanism you had not considered, or a common cause that is missing from both?

3. **Which edges does your theory include that PC removes?** Look up the relevant test in `localTests()` output. Is the association truly absent, or merely too small to detect at this sample size?
4. **Where PC points an arrow, do you believe it?** Check whether the direction is substantively possible. An implausible arrow can be suggestive of an assumption violation.
5. **Where PC leaves an edge undirected, what would settle it?** Can we infer it using time order, an experiment, or an additional variable?

! What PC assumes

Several assumptions are needed to infer the causal graph from statistical associations alone. With any causal discovery algorithm, the result is only as good as these assumptions, and with real-world data, some may likely be violated.

- **Causal Markov Condition**: Whatever the graph blocks really is statistically unrelated in the data.
- **Faithfulness**: Whatever is statistically unrelated in the data really is blocked in the graph. In practice, this implies that no two causal paths cancel each other out exactly. For instance, a positive direct effect offset by a negative indirect effect could look like no effect at all.
- **No hidden confounders**: every common cause of the measured variables is itself measured.
- **No cycles in the causal graph**: causation runs one way through the graph, and nothing returns to itself.
- **Linearity and normality (because of independence test chosen)**: `gaussCItest` reads independence off partial correlations, which capture *linear* relations only.

7.6 Concluding thoughts

Throughout this tutorial, we have learnt that the graph produced by the PC algorithm cannot be taken, at face value, as the "true" causal structure. A discovered graph is an estimate that is built on assumptions you can substantively consider but mostly cannot check empirically. Expert knowledge still plays an important role and is needed to interpret the output, informing us when the model output simply cannot be true (e.g., teacher pressure influencing age) or when core assumptions are likely violated. Causal discovery is an exploratory tool that can, under certain conditions, provide us useful information in formulating theories (e.g., do our theory's predictions allign with the data?) and help us think about what experiments to prioritize. But perhaps most importantly, these causal tools force us to make the assumptions about causality (in our theories) explicit.

Think back on last week's machine learning analyses, for example. Do you think that all of the important predictors of the outcome you chose are in fact causes of it? Or is it possible that some of the predictors are *caused by* your chosen outcome, or correlated with it for irrelevant reasons?

7.7 Homework

Apply the causal discovery pipeline from this tutorial (Part 3) to the dataset you prepared for last week's machine learning analysis:

1. Write down your theoretical DAG, and list the conditional independencies it implies.
2. Test those implications with `localTests()`.
3. Run the PC algorithm on the same data, and compare the two graphs.

Interpret the results you've obtained, and reflect on your previous theory (the DAG). Do you see any causal links that are not in your theory-based DAG but should be? Did you learn anything new about the direction of effects? Are any of the links counterintuitive or wrong? Which assumption do you think is responsible? Write your reflections down to become part of your final report.

8 FAIR Theory

8.1 Weekly Readings

- Van Lissa, C. J., Peikert, A., Ernst, M. S., van Dongen, N. N. N., Schönbrodt, F. D., & Brandmaier, A. M. (2026). T o Be FAIR: Theory Specification Needs an Update. Perspectives on Psychological Science, 17456916251401850. https://doi.org/10.1177/17456916251401850

8.2 Tutorial

This tutorial helps you make your theory FAIR. You will find an R-script for the steps described in this tutorial in the file `fair_theory.R`, in the `patterns_principles_exercises` folder. Use one computer to complete this tutorial together as a group. You will need to install some software in the process, so it's easier to just do that on one system.

```
library(theorytools)
```

8.2.1 Learning Goals

- ☐ Create a project folder for your theory
- ☐ Put this folder under version control with 'Git'
- ☐ Connect the local repository to a remote ('GitHub') repository
- ☐ Add a shareable theory file to the repository
- ☐ Add a LICENSE file to the repository (we recommend CC0)
- ☐ Add a README file to the repository
- ☐ Add a 'Zenodo' metadata to the repository
- ☐ Push these changes to the remote repository
- ☐ Turn on 'Zenodo' archiving for the remote repository
- ☐ Publish a release of your theory
- ☐ Verify that 'Zenodo' mints a DOI for your theory and its latest release

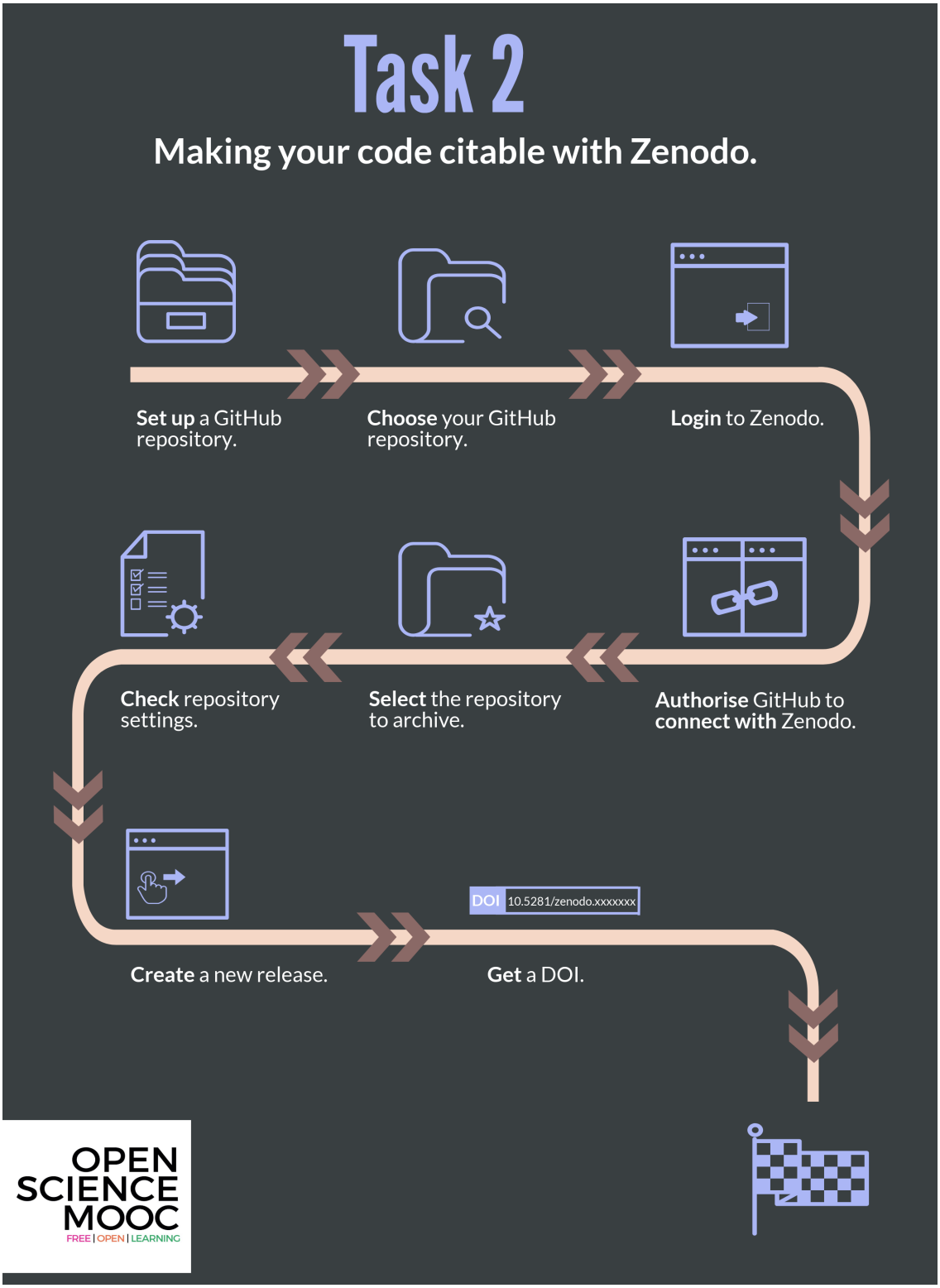


Figure 8.1: Conceptual workflow for this task.

8.2.2 Running Example: SDT

As an example, we will use a specification of Self-Determination Theory (SDT). Here, we use a DAG - but you can archive anything you like, although it is recommended to use plain text files, which are machine- and human-readable without special software. Note that I've added an example of specifying a functional form for one of the links in the DAG; you can see more examples of this kind of annotation here.

```
theory <- '
dag {
external_event
healthy_development
integration
intrinsic_motivation
locus_of_causality
needs
wellbeing
external_event -> intrinsic_motivation [form="external_event:needs"]
external_event -> locus_of_causality
external_event -> needs
integration -> healthy_development
integration -> wellbeing
intrinsic_motivation -> healthy_development
intrinsic_motivation -> wellbeing
locus_of_causality -> intrinsic_motivation
needs -> integration
needs -> intrinsic_motivation  [form="external_event:needs"]
needs -> wellbeing
}'
```

8.2.3 Creating a Project Folder

Create an empty project folder, which will become the theory archive. If we want to create a new folder called `sdt` in the existing folder `~/theories/`, we can call:

```
project_path <- file.path("~/theories", "sdt")
dir.create(project_path)
```

8.2.4 Adding a Shareable Theory File to the Repository

Your theory should be represented as a digital artifact, such as a structured plain-text document or a machine-readable file (e.g., 'DOT', 'JSON', 'YAML', 'R' code). You can simply write the text vector above to a plain text file, like so:

```
writeLines(theory, file.path(project_path, "theory.txt"))
```

8.2.5 Documenting Reusability with a LICENSE

A license ensures that others know how they can legally reuse your work. We recommend a CC0 (Creative Commons Zero) license for FAIR theory, which waives all copyright protection and places your work in the public domain. This recommendation is based on the fact that copyright protection does not cover ideas, and the assumption that FAIR theory is created to maximize reuse. Other licenses are available, see https://choosealicense.com. You can add a license file to your repository like so:

```
worcs::add_license_file(path = project_path, license = "cc0")
```

8.2.6 Documenting Interoperability with a README File

A README file describes the repository's contents and purpose, making it easier for others to understand your theory's potential for interoperability and reuse. The `theorytools` package contains a function to generate a README file with appropriate sections for FAIR theory, which can be used like so:

```
theorytools::add_readme_fair_theory(title = "Self-Determination Theory",
                                    path = project_path)
```

We encourage users to edit the resulting `README.md` file, in particular, to add relevant information about X-interoperability. For guidance on writing a README file for theory, see this vignette.

8.2.7 Adding 'Zenodo' Metadata to the Repository

In a later step, we will archive the theory on 'Zenodo'. Creating a `.zenodo.json` file with metadata about your theory allows the project to be indexed automatically. This can be done by running:

```r
add_zenodo_json_theory(
  path = project_path,
  title = "Self-Determination Theory",
  keywords = c("motivation")
)
```

8.2.8 Version Controlling The Project Folder

We will use 'Git' to version control the project folder. This means that changes to the files are explicitly tracked, and the entire history of the theory-project is logged. One student in your group should install 'Git' on their computer, as explained here. You can verify that 'Git' is installed and working by running:

```r
worcs::check_git()
```

If this function shows a green checkmark, you can initialize version control in your project repository by running:

```r
gert::git_init(path = project_path)
```

8.2.9 Connecting to a Remote ('GitHub') Repository

To make your FAIR theory accessible to collaborators and discoverable by the wider community, you must connect your local 'Git' repository to a remote repository on a platform like 'GitHub'.

Before proceeding, ensure that one of your group members has a 'GitHub' account. To authorize 'R' to interact with your 'GitHub' account, run `usethis::create_github_token()`, which takes you to a website to create a personal access token (PAT). Copy it, then run `gitcreds::gitcreds_set()` and paste the PAT when asked. If you still experience problems try `usethis::gh_token_help()` for help.

To check that you are ready to proceed, run:

```r
worcs::check_github()
```

If you see a green checkmark, you can create a new repository on 'GitHub' directly from 'R':

```r
worcs::git_remote_create("fair_sdt", private = FALSE)
```

This command will create a new public repository on 'GitHub' and link it to your local repository. The `private = FALSE` argument ensures the repository is public by default.

Either way, assuming the name of that repository is `fair_sdt`, you can connect it to your project folder as follows:

```
worcs::git_remote_connect(project_path, remote_repo = "fair_sdt")
```

8.2.10 Pushing These Changes to the Remote Repository

Version control requires adding files to be tracked to the repository. The `worcs` function `worcs::git_update()` does this automatically, acting like a kind of "quick-save" or "save all" function:

```
worcs::git_update("First commit of my theory", repo = project_path)
```

8.2.11 Check Your 'GitHub' Repository

Navigate to your repository on 'GitHub' and check that all committed files, including the theory file, license, README, and 'Zenodo' metadata, are now visible in the remote repository (green box in the image below).

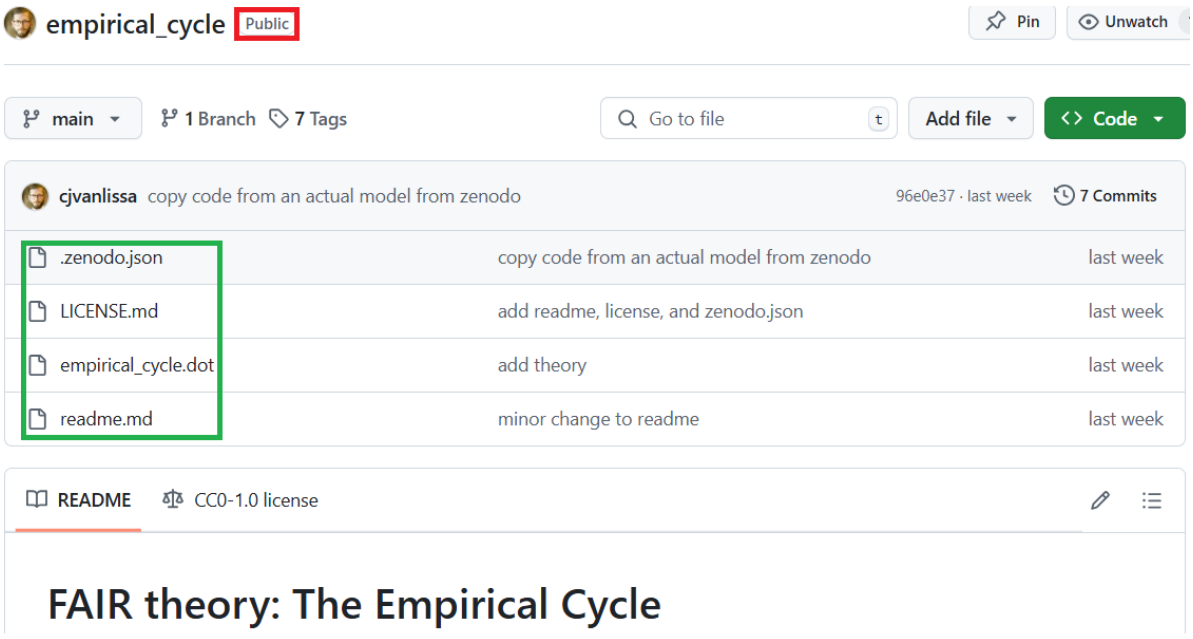


Figure 8.2: Front Page of a 'GitHub' Repository

Furthermore, the repository visibility must be set to "Public" to ensure that 'Zenodo' can discover and archive it. If you created the repository programmatically as shown above, it should already be public (see red box in the image above).

8.2.12 Login to 'Zenodo'

Head over to zenodo.org. 'Zenodo' is a platform where you can permanently archive your code and other project elements. 'Zenodo' does this by assigning projects a **Digital Object Identifier** (DOI), which also helps to make the work more citable. This is different to 'GitHub', which acts as a place where the actual work on a project takes place, rather than long-term archiving of it. At 'GitHub', content can be modified, deleted, rewritten, and irreversibly changed, which makes it a bit concerning to be used for longer lasting referencing purposes. 'Zenodo' offers more security and permanence for research outputs.

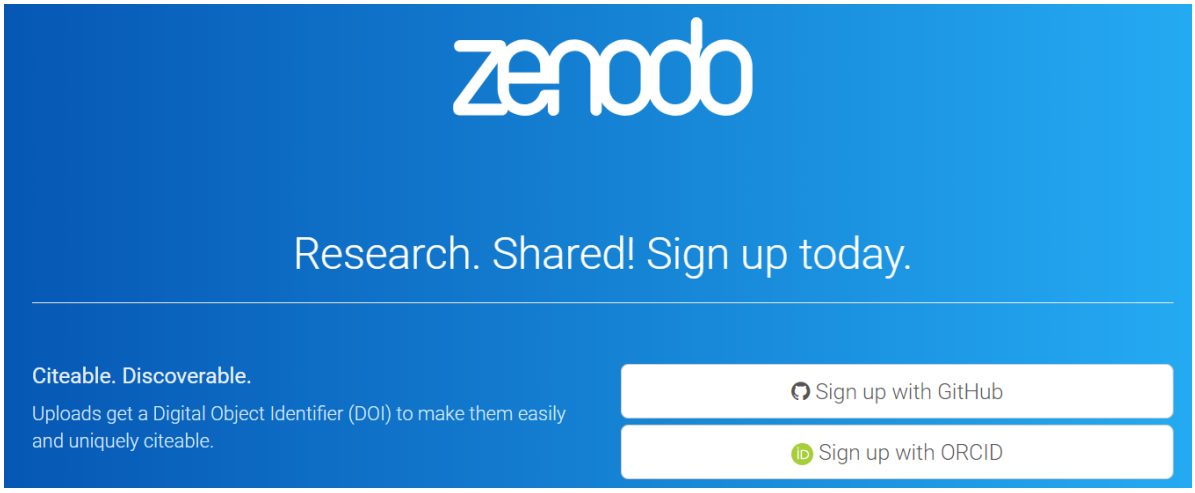


Figure 8.3: Sign up for 'Zenodo'

One student should create a 'Zenodo' account. You can **login using your 'GitHub' account** to make the registration process easy.

8.2.13 Authorize 'GitHub' to connect with 'Zenodo'

On the 'Zenodo' website authorize it to connect to your 'GitHub' account in the 'Using 'GitHub'' section. Here, 'Zenodo' will redirect you to 'GitHub' to ask for permissions to use 'webhooks' on your repositories. You want to authorize 'Zenodo' here with the permissions it needs to form those links.

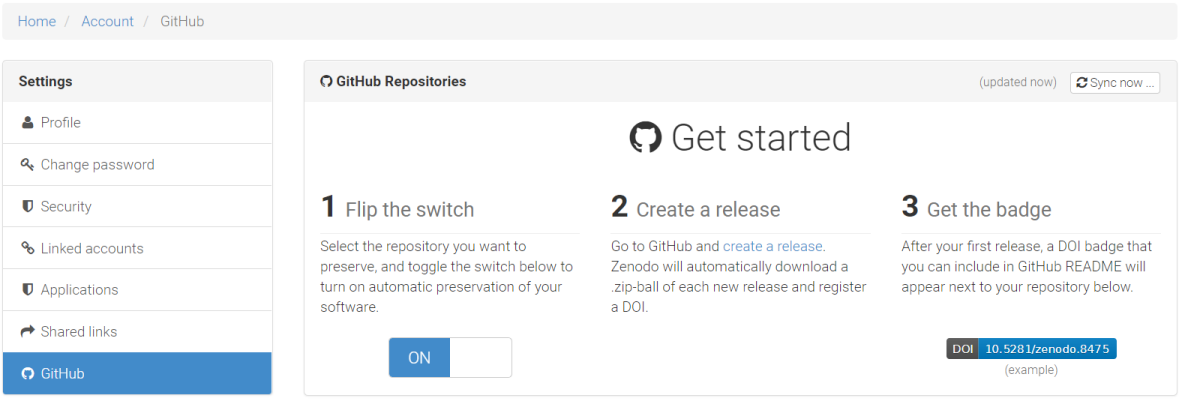


Figure 8.4: Authorize to connect with 'GitHub'

8.2.14 Select the Repository to Archive

> **Use the Zenodo Sandbox**
>
> You will now archive your project. This costs Zenodo some money and storage space. To be considerate, we will not **really** archive our project, but instead just practice archiving it in a "sandbox" environment.
> You can access the sandbox environment by navigating to http://sandbox.zenodo.org/. The links below already use the sandbox environment. If you ever need to properly archive something in the future, simply remove `sandbox.` from the link addresses.

If you have got this far, this means that 'Zenodo' is now authorized to configure the repository webhooks that it needs to archive the repository and issue it a DOI. To do this, on the 'Zenodo' website navigate to the 'GitHub' repository listing page and simply "flip the switch" next to your repository. If your repository **does not show up** in the list, you may need to press the 'Synchronize now' button. At the time of writing, we noticed that it can take quite a while (hours?) for 'Zenodo' to detect new 'GitHub' repositories. If so, take a break or come back to this last step tomorrow!

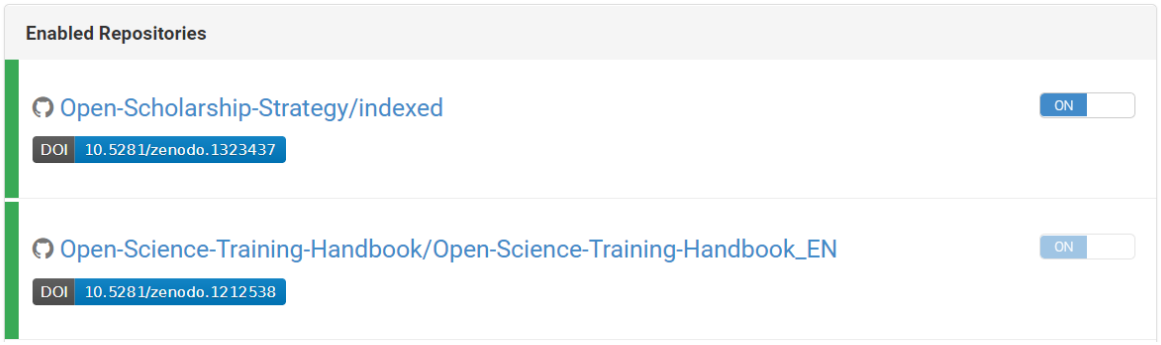


Figure 8.5: Enable individual 'GitHub' repositories to be archived in 'Zenodo'

8.2.15 Create a New Release

To archive a repository on 'Zenodo', you must create a new release. You can do this using the following code:

```
worcs::git_release_publish(repo = project_path)
```

If you have not previously published any releases, this function will assume that you want to use semantic versioning for both the release tag and the release title. This means that the first release will be labeled with version number "0.1.0". Each subsequent release will automatically increment the trailing digit, i.e.: "0.1.1", "0.1.2". If you make a major change to the theory, you may want to manually increment the middle digit like so:

```
worcs::git_release_publish(repo = project_path,
                           tag_name = "0.2.0",
                           release_name = "0.2.0")
```

8.2.16 Verify on 'Zenodo'

To verify that your release was archived on 'Zenodo' and assigned a DOI, you need to visit the Uploads tab.

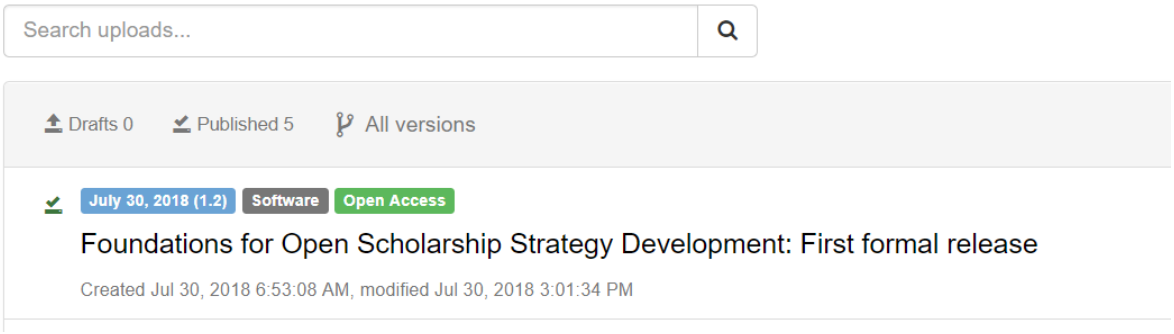


Figure 8.6: Check the new release has been uploaded.

8.2.17 Updating Meta-Data

We can further document our 'Zenodo' archive as a FAIR theory by adding some extra information on 'Zenodo'. Note that, if you created a `.zenodo.json` file in a previous step, some of these metadata will be populated automatically. On 'Zenodo' click the Upload tab in the main menu, where you should find your newly uploaded repository.

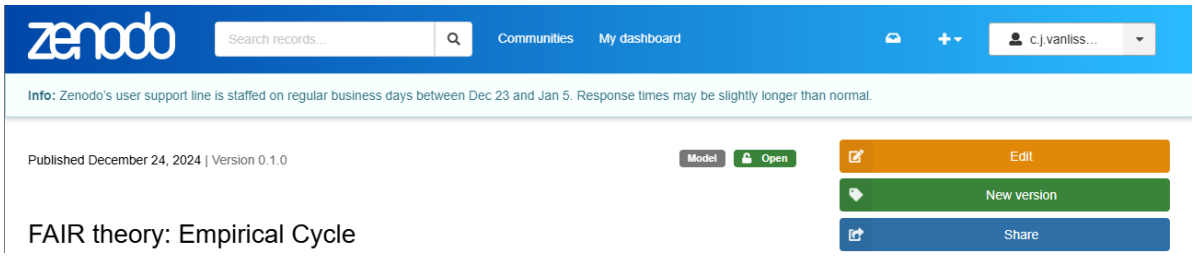


Figure 8.7: Click the orange Edit button.

Click the orange `Edit` button, and verify/supply the following information:

- ☐ Resource type: Should be set to `Model`
- ☐ Title: Should be prefaced with `FAIR theory:`
- ☐ Keywords and subjects: The first keyword should be `fairtheory`
- ☐ Related works: Add the DOIs/identifiers of related (print) works. Use the `Relation` field as appropriate. For example:
  - `Is documented by` a theory paper you wrote, in which you introduce this FAIR theory

- `Is derived from` an existing theory, which was published in print (paper, book chapter) but not made FAIR

☐ References: Optionally, cite related works in plain text. For example, here we can provide the citation for the SDT theory book:

> Ryan, R. M. (Ed.). (2023). The Oxford Handbook of Self-Determination Theory (1st ed.). Oxford University Press. https://doi.org/10.1093/oxfordhb/9780197600047.001.0001

To save these changes, click 'Publish'.

8.2.18 Verifying That 'Zenodo' Mints a DOI for Your Theory

After publishing a release, 'Zenodo' will archive the repository and mint a DOI. Verify this by checking the 'Zenodo' entry for your repository, where the DOI will be displayed. Include this DOI in any citations or references to your theory to enhance its discoverability and reusability.

The 'GitHub'/'Zenodo' integration will assign one "mother-DOI" to the project, which will always resolve to the latest version, as well as a unique DOI to each version/release of the FAIR theory. This enables users to refer to and cite either the theory in general or specific versions of the theory. The list of authors for the citation is automatically determined by the 'GitHub' user account names used by the repository - this can be edited on 'Zenodo', as explained above. DOIs used in 'Zenodo' are registered through the DataCite service.

> **Pro-tip**: Check the `Citation` field on the 'Zenodo' page, and copy-paste it into the README file of your 'GitHub' repo to make cross-linking even easier (or refer users to the 'Zenodo' page to find the citation, which obviates the need to manually update this information). Click the DOI badge in the `Details` field to get instructions on how to add a clear highlighted DOI badge to your 'GitHub' repository, for users to see and make use of your DOI:

8.2.19 CONGRATULATIONS!

Your FAIR theory is now archived in 'Zenodo', and with a DOI that can be versioned to reflect updates to the repository version through time. You should be able to see details of this on the 'GitHub' 'Zenodo' page for your repository. This also means that your archived projects can get picked up by other indexing services and search engines that use DOIs too.

Providing a long-term archive and a DOI for your work is required for others to be able to properly cite it, as this provides basic citation metadata. For Open Science, it is important to be able to comprehensively cite the resources that you use in your research, including theory, and this workflow enables that to happen, in line with best practices. Making theory FAIR

also helps elevate the standard of theory to that of the standard of other research outputs, like papers and software.

8.2.20 Checklist for citing your project

So now you have a sustainably archived 'GitHub' repository in 'Zenodo' that is ready to be re-used and cited! Before continuing, make sure that you have:

- ☐ Linked your 'GitHub' project to 'Zenodo'. If you see a complete copy of your 'GitHub' repository in 'Zenodo' then things are working.
- ☐ 'Zenodo' and 'GitHub' integrated setup works nicely. For example have all the author names, and correct project title come across to 'Zenodo'. If not, or if authors just have nicknames you can edit these details in 'Zenodo'.
- ☐ Project has a first release, with a DOI. You should have a DOI displayed on your projects 'Zenodo' page. This first DOI is called the 'concept DOI' and is the master DOI linking to all subsequent release DOIs. Copy this DOI link and embed it in your 'GitHub' projects README page. You're done!

8.3 Homework

Complete this tutorial if you did not manage to do so during the lab, and work on your final theory and report.

References

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A Setup your Computer

This is a tutorial on how to set up your personal computer for the course. It guides you through the installation of several software packages. This vignette does not assume a prior installation of `R`, so it is suitable for novice users. You only have to perform these steps once for every computer you intend to use, and the entire process should take approximately 30 minutes if you start from scratch. In case some of the software is already installed on your system, you can skip those related steps.

Follow these steps in order:

1. Install R (free)
2. Install 'RStudio' Desktop (Free)
3. Download the RStudio project for this course into a folder where you'd like to keep your working files for assignments.
4. Unzip the compressed folder, open it, and open `patterns_principles_exercises.Rproj` in RStudio.
5. Once RStudio has loaded, you' ll see a panel in the bottom right labeled "Files". Click the file named `setup.R`. It will load. Select all code in this file, then either press [CONTROL+ENTER], or click the `Run` button.